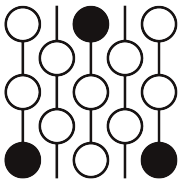
Review of Problem 7 – ProofCouncil

First Proof Third Batch
Reviewer 1
October 8–9, 2026



1 Review

Correctness.

- 1: essentially flawless mathematical argument
- 0.5: minor errors that are easily corrected
- 0: blank solution, or acknowledged partial result
- −0.5: major errors
- −1: proposed proof makes no progress towards the result.

Note that points should be docked for wrong statements/ solutions.
Score: 0.5

Exposition.

- 1: well-written and clear solution.
- 0: argument that is possible to follow with serious effort
- −1: very poor exposition

Score:-1

Attributions.

- 1: solutions contains fully appropriate attributions to related work and theorems being used.
- 0: moderate issues with missing attributions or hallucinated references
- −1: blatant plagiarism or hallucinations

Score: 1

> **Reviewer 1**
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> A note on attributions: no need to cite both [HK] and [HKa]; these are journal and preprint versions of the same paper and the journal version is comprehensive

Reviewer 1

General review: Was able to determine that this one is correct, but it took serious effort because the exposition is at a level that is impossible to grade in two days. The impossibility of grading in this time frame strictly without use of AI applies to all four submissions. Lots of convoluted computations. Diagrams are absolutely necessary for this sort of paper and it seems like a lot of these arguments can be compactified based on the arguments in some of the references.

2 Submission with inline comments

Abstract

Let f be a quadratic rational map one of whose critical points lies in a cycle A of length four and whose other critical value v lies outside A, and let $B = A \cup \{v\}$. Epstein's deformation space $\mathcal{D}_f \subset \mathcal{T}(\mathbb{S}^2, B)$ is known to be disconnected (Hironaka–Koch). We write down an explicit simple closed curve γ on $(\mathbb{S}^2, B)$, for a normalized model map the image of a round circle under two Dehn twists about a second round circle, transported to a general f along a path in parameter space, and prove that the Dehn twist T_γ itself preserves $\mathcal{D}_f$ and that the translates $T_\gamma^n(D_0)$, $n \in \mathbb{Z}$, of the base component D_0 are pairwise distinct components, using the covering-space framework of Hironaka–Koch and a winding-number obstruction.

3 Introduction

3.1 Deformation spaces

Throughout, $\mathbb{P}^1$ is the Riemann sphere, which we also regard as the oriented topological sphere $\mathbb{S}^2$. For a finite set $B \subset \mathbb{P}^1$ with $|B| \geq 3$, the *Teichmüller space* $\mathcal{T}(B) = \mathcal{T}(\mathbb{S}^2, B)$ is the set of orientation-preserving homeomorphisms $\phi\colon \mathbb{P}^1 \to \mathbb{P}^1$ (*markings*), where $\phi \sim \phi'$ when $\phi' = \mu \circ \phi \circ h$ for a Möbius transformation μ and a homeomorphism h isotopic to the identity relative to B. Pulling back the standard complex structure by ϕ identifies $\mathcal{T}(B)$ with the complex structures on $(\mathbb{S}^2, B)$ up to isotopy relative to B; the class $\tau_0 = [\mathrm{id}]$ is the *base point*.

Let $f\colon \mathbb{P}^1 \to \mathbb{P}^1$ be an orientation-preserving branched cover of degree $d \geq 2$ with set of critical values $V(f)$, and let $A, B \subset \mathbb{P}^1$ be finite sets with $|A|, |B| \geq 3$ and $f(A) \cup V(f) \subset B$. Pulling back the complex structure of a marking ϕ under f gives a complex structure on $(\mathbb{S}^2, A)$; concretely, there are a homeomorphism ψ, unique up to post-composition with a Möbius transformation, and a rational map F with

$$F \circ \psi = \phi \circ f, \tag{3.1}$$

and $f^*[\phi] := [\psi] \in \mathcal{T}(A)$. This is the Thurston pullback map $f^*\colon \mathcal{T}(B) \to \mathcal{T}(A)$; it is holomorphic [BCT, §1.3, Prop. 1.4].

> **Reviewer 1**
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> For the last reference, Can also cite Douady–Hubbard, "A proof of Thurston's topological characterization of rational functions," Acta Math. 171 (1993)

We call ϕ the *target marking* and ψ the *source marking* of the diagram (3.1). If $A \subset B$, forgetting the points of $B \setminus A$ gives the holomorphic *forgetful map* $i_*\colon \mathcal{T}(B) \to \mathcal{T}(A)$, $[\phi] \mapsto [\phi]$. When $f(A) \cup V(f) \cup A \subset B$, both maps are defined, and the *deformation space* of f is their equalizer

$$\mathcal{D}_f = \{\tau \in \mathcal{T}(B) : f^*(\tau) = i_*(\tau)\}.$$

This notion is due to Epstein [Ep]; see [BCT, §1.4] for an exposition. When f is rational, as in the problem considered here, we may take $\psi = \mathrm{id}$, $F = f$ in (3.1) for $\phi = \mathrm{id}$, so $\tau_0 \in \mathcal{D}_f$. (By Epstein's theorem [BCT, Thm. 1.5], under the standing hypotheses above, if f is not a topological $(2, 2, 2, 2)$-map in the sense of [BCT, §1.4], then $\mathcal{D}_f$, whenever nonempty, is a smooth complex submanifold of $\mathcal{T}(B)$ of dimension $|B \setminus A|$. For the rational family considered here $\tau_0 \in \mathcal{D}_f$, and the periodic critical point excludes the exceptional type.)

The *pure mapping class group* $\mathrm{PMod}(B)$, the group of isotopy classes relative to B of orientation-preserving homeomorphisms of $\mathbb{P}^1$ fixing B pointwise, acts on $\mathcal{T}(B)$ on the right by

$$[\phi] \cdot h = [\phi \circ h]; \tag{3.2}$$

for a subset $D \subset \mathcal{T}(B)$ we write $h(D) = \{\tau \cdot h : \tau \in D\}$. A mapping class that preserves $\mathcal{D}_f$ permutes its connected components. For a simple closed curve γ in $\mathbb{P}^1 \setminus B$ we write $T_\gamma \in \mathrm{PMod}(B)$ for the Dehn twist about γ. The question addressed here asks for an explicit γ such that some power of T_γ preserves $\mathcal{D}_f$ and has an infinite orbit on the components.

3.2 The family and the result

Let f be a quadratic rational map with critical points c_1, c_2 such that c_1 lies in a cycle of length 4 and the critical value $v = f(c_2)$ is not in this cycle, and put

$$A = \{c_1, f(c_1), f^{\circ 2}(c_1), f^{\circ 3}(c_1)\}, \qquad B = A \cup \{v\}.$$

Then $f(A) = A$ and $V(f) = \{f(c_1), v\} \subset B$, so $\mathcal{D}_f \subset \mathcal{T}(B)$ is defined; $\mathcal{T}(B)$ has dimension 2. Conjugating f by the Möbius transformation that sends $c_1, f(c_1), f^{\circ 2}(c_1)$ to $0, \infty, 1$ normalizes the cycle to $0 \mapsto \infty \mapsto 1 \mapsto s \mapsto 0$, and Section 5 shows that this forces f to become

$$F_s(z) = 1 + \frac{-s^2 - s + 1}{(s-1)z} + \frac{s(2s-1)}{(s-1)z^2}, \qquad s \in U, \tag{3.3}$$

for an explicit Zariski-open set $U \subset \mathbb{C}$ given in (5.2). Our model is F_a with $a = 3/5$, whose marked set is $B_a = \{0, \infty, 1, a, v_a\}$, where $v_a = 121/120$ is its second critical value (computed in (5.1)).

The answer (Theorem 5.2) is as follows. Let C be the round circle $|z| = 241/240$, which separates $\{\infty, v_a\}$ from $\{0, 1, a\}$, and let η be the round circle $|z - 4/5| = 1/4$, which encloses exactly $a, 1, v_a$. Put $\gamma_a = T_\eta^2(C)$, the image of C under two Dehn twists about η (with the sign of Convention 4.1; either sign works, Remark 9.4). For a general f of the family, let γ be obtained from γ_a by following any path in U from a to the parameter of f and dragging the marked points along. Then T_γ preserves $\mathcal{D}_f$, and the translates $T_\gamma^n(D_0)$, $n \in \mathbb{Z}$, of the component $D_0 \ni \tau_0$ are pairwise distinct components; so the power of the twist can be taken to be 1. (A path is needed: γ_a and C separate B_a in the same way, so separation data do not determine the isotopy class of γ.)

3.3 Relation to earlier work

Hironaka and Koch [HK] proved that for this family $\mathcal{D}_f$ has infinitely many connected components ([HK, Thm. 1.4]), introducing the tools we use: the surface of marked quadratic maps with its degree-four covering of moduli space and its diagonal ([HKa, §§4.1–4.2], [HK, §3]), the description of $\mathcal{D}_f$ through mapping classes that lift under f and through equalizers ([HK, §§2.8–2.11]), and a symmetric quotient of the parameter surface used as an obstruction ([HK, §2.12]); they also exhibit a mapping class preserving $\mathcal{D}_f$ with an infinite orbit on components ([HK, Prop. 3.2]). Here [HKa] is the first arXiv version of their paper and [HK] the revised author version, whose numbering we follow. Firsova, Kahn and Selinger [FKS] treat the same family ([FKS, §6]; numbering of arXiv v4), study twists along multicurves adapted to the pullback ("equalizing multicurves"), and prove that these deformation spaces are not contractible ([FKS, Lemma 8.4]). The present note adds the identification of a single, explicitly drawn Dehn twist with an infinite component orbit, which we have not found in [HK] or [FKS], with an elementary verification; our obstruction map Φ (Section 9) is a form of the symmetric quotient of [HK].

3.4 Strategy and outline

The proof has five steps. (1) Section 6 factors the universal cover $\mathcal{T}(B_a) \to M$ of the moduli space M of five marked points through a finite cover $\mathcal{W} \to M$, the surface of quadratic maps with the prescribed portrait of marked orbits; the pullback and forgetful maps become the coordinate projections of $\mathcal{W}$, and $\mathcal{D}_{F_a}$ lies over the diagonal $\mathcal{V} \subset \mathcal{W}$. (2) Section 7: a loop g in $\mathcal{W}$ acts on $\mathcal{T}(B_a)$ as a mapping class, which preserves $\mathcal{D}_{F_a}$ if the two coordinate loops of g are homotopic, and preserves the base component D_0 if and only if g is homotopic in $\mathcal{W}$ to a loop in $\mathcal{V}$. (3) Section 8 exhibits a loop $g = \ell^{-1} m \ell$ in $\mathcal{W}$ acting as $T_{\gamma_a}^{\pm 1}$: m is a meridian acting as the twist about C, and the point push along ℓ is identified exactly by a Joukowski substitution. (4) Section 9 shows that no power g^n, $n \neq 0$, is homotopic to a loop in $\mathcal{V}$, using a map $\Phi = (t, L) \colon \mathcal{W} \to (\mathbb{C} \setminus \{0,1\}) \times \mathbb{C}^*$ whose first coordinate becomes, on an enlarged diagonal, the two-sheeted covering $w \mapsto w^2$, while its second coordinate imposes a linear relation between two winding numbers that the loops g^n violate. (5) Section 10 transports the conclusion to every map of the family by lifting an isotopy of marked points through the branched covers.

> **Reviewer 1**
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> Above needs pictures, it's next to impossible to understand.

4 Conventions on loops, point pushing and twists

For B as above, the *moduli space* $\mathcal{M}(B)$ is the set of injective maps $B \to \mathbb{P}^1$ modulo post-composition with Möbius transformations. Choosing three points of B and normalizing their images to $0, \infty, 1$ identifies $\mathcal{M}(B)$ with an open subset of $\mathbb{C}^{|B|-3}$. The projection $\pi_B \colon \mathcal{T}(B) \to \mathcal{M}(B)$, $[\phi] \mapsto [\phi|_B]$, is a holomorphic universal covering whose deck group is $\mathrm{PMod}(B)$ acting by (3.2); see [KPS, §2.1].

Two moduli spaces occur. For four marked points labelled $0, \infty, 1, x$, and for five points labelled $0, \infty, 1, y, z$, we use respectively

$$Y = \mathbb{C} \setminus \{0,1\}, \qquad M = \{(y,z) \in Y^2 : y \neq z\}. \tag{4.1}$$

Thus for $A \supset \{0, \infty, 1\}$ with $|A| = 4$, $\pi_A \colon \mathcal{T}(A) \to Y$ records the fourth point under the marking normalized to fix $0, \infty, 1$, and similarly $\pi_B \colon \mathcal{T}(B) \to M$ when $|B| = 5$.

Products of based loops are read from right to left: $\alpha\beta$ traverses β first and then α, matching composition of homeomorphisms. For a closed loop ω in $\mathbb{C} \setminus \{c\}$ we write $\mathrm{wind}_c(\omega) \in \mathbb{Z}$ for its winding number about c; for a non-closed path, $\mathrm{wind}_c(\omega)$ denotes the total change of argument of $\omega - c$ divided by 2π, which is additive under concatenation and changes sign under reversal.

Let δ be a loop in $\mathcal{M}(B)$ based at $\pi_B(\tau_0)$, that is, a motion of the marked points. By isotopy extension there is an ambient isotopy h_t of $\mathbb{P}^1$ with $h_0 = \mathrm{id}$ such that $h_t|_B$ follows $\delta(t)$. The mapping class $P(\delta) = [h_1] \in \mathrm{PMod}(B)$ depends only on the homotopy class of δ, and $[h_t]$ is the lift of δ to $\mathcal{T}(B)$ starting at τ_0, with endpoint $\tau_0 \cdot P(\delta)$. Consequently the deck transformation associated with δ, the one sending τ_0 to the endpoint of this lift, is

$$[\phi] \longmapsto [\phi \circ P(\delta)]. \tag{4.2}$$

Point pushing respects products, $P(\alpha\beta) = P(\alpha)P(\beta)$, since the isotopy following $\alpha\beta$ may be taken to be h_t^{β} followed by $h_t^{\alpha} \circ h_1^{\beta}$. Because (4.2) is a right action, deck transformations of products compose in the opposite order; we only use deck transformations of powers of a single loop. We write $g(\tau)$ for the deck transformation (4.2) of a loop g applied to τ.

When only one marked point v moves, δ is a loop ζ in $\mathbb{P}^1 \setminus (B \setminus \{v\})$ based at v, and $P(\zeta)$ is the *point-pushing map* $\mathrm{Push}(\zeta)$ of the Birman exact sequence [FM, §4.2, Thm. 4.6]. We use two standard facts. (P1) If ζ is a simple loop, then $\mathrm{Push}(\zeta) = T_{\zeta_1} T_{\zeta_2}^{-1}$, where ζ_1, ζ_2 are the two boundary curves of a regular neighbourhood of ζ, the order being fixed by the orientation of ζ and the sign convention for twists [FM, Fact 4.7]. (P2) For every mapping class h and simple closed curve c, $hT_c h^{-1} = T_{h(c)}$ [FM, Fact 3.7].

Two consequences of (P1) are used. First, if $\Delta \subset \mathbb{P}^1$ is a closed disk with $\partial\Delta \cap B = \emptyset$ and $\Delta \cap B = \{v, w\}$, and ζ is a simple loop of v in $\Delta \setminus \{w\}$ going once around w, then one boundary curve of its regular neighbourhood bounds a disk containing only w (trivial twist) and the other is isotopic to $\partial\Delta$, so

$$\mathrm{Push}(\zeta) = T_{\partial\Delta}^{\pm 1}. \tag{4.3}$$

Second, if κ, η are disjoint simple closed curves bounding an annulus meeting B only in v, with κ inside η in a given disk picture, and ζ is a simple loop of v once counterclockwise around the core of the annulus, then $\mathrm{Push}(\zeta) = T_\eta^{\epsilon} T_\kappa^{-\epsilon}$ with $\epsilon = \pm 1$ depending only on the sign convention for Dehn twists.

Convention 4.1. We fix the sign convention for Dehn twists so that $\epsilon = +1$, that is, $\mathrm{Push}(\zeta) = T_\eta T_\kappa^{-1}$ for such a counterclockwise circuit. With the opposite convention the boxed formula (5.5) of Section 5 reads $T_\eta^{-2}(C)$; both curves have the stated property (Remark 9.4).

5 The model map and the curve

Let f be as in Section 3.2 and conjugate it so that its critical cycle is $0 \mapsto \infty \mapsto 1 \mapsto s \mapsto 0$ with 0 the critical point. A quadratic rational map with a critical point at 0 mapping to ∞ has a double pole at 0 and no other pole, so it is $1 + b_s/z + c_s/z^2$ after using $F(\infty) = 1$. Solving the linear equations $F(1) = s$ and $F(s) = 0$ for b_s, c_s gives $b_s = (-s^2 - s + 1)/(s-1)$ and $c_s = s(2s-1)/(s-1)$, which is (3.3). The other critical point of F_s is $-2c_s/b_s$, and its critical value is $1 - b_s^2/(4c_s)$, which simplifies to

$$v(s) = -\frac{(s^2 - 3s + 1)^2}{4s(s-1)(2s-1)}. \tag{5.1}$$

The map has degree two and the four cycle points are distinct exactly when $s \neq 0, 1, 1/2$ (for $s = 1/2$ one has $c_s = 0$). The identities

$$v(s) - 1 = -\frac{(s^2 + s - 1)^2}{4s(s-1)(2s-1)}, \qquad v(s) - s = -\frac{(3s^2 - 3s + 1)^2}{4s(s-1)(2s-1)}$$

show that $v(s) \notin \{0, \infty, 1, s\}$, that is, $v(s)$ lies outside the cycle, exactly when the three quadratic factors below are nonzero. So the maps in the problem are, up to Möbius conjugacy, precisely the F_s with

$$s \in U = \left\{ s \in \mathbb{C} : s(s-1)(2s-1)(s^2 - 3s + 1)(s^2 + s - 1)(3s^2 - 3s + 1) \neq 0 \right\}. \tag{5.2}$$

The set U, a cofinite subset of $\mathbb{C}$, is connected. We write $A_s = \{0, \infty, 1, s\}$ and $B_s = A_s \cup \{v(s)\}$ for the marked sets of F_s. Our model is the parameter $a = 3/5$:

$$F_a(z) = 1 - \frac{1}{10z} - \frac{3}{10z^2}, \qquad 0 \longmapsto \infty \longmapsto 1 \longmapsto a \longmapsto 0, \tag{5.3}$$

with other critical point -6 and other critical value $v_a = v(3/5) = 121/120$. On the sphere marked by $B_a = \{0, \infty, 1, a, v_a\}$ draw the round circles

$$C = \left\{|z| = \tfrac{241}{240}\right\}, \qquad \eta = \left\{\left|z - \tfrac{4}{5}\right| = \tfrac{1}{4}\right\}. \tag{5.4}$$

The exterior of C contains exactly the marked points ∞, v_a (note $v_a = 242/240$), and the interior of η contains exactly $a, 1, v_a$ (the distances from $4/5$ to a, 1, v_a are $1/5$, $1/5$, $25/120$, all less than $1/4$, while $|0 - 4/5| > 1/4$). The two circles intersect. Our model curve is

$$\boxed{\gamma_a = T_\eta^2(C),} \tag{5.5}$$

with the sign of Convention 4.1.

Construction 5.1 (Transport to a general map)**.** *Let f be a map of the family with normalized form F_s, $s \in U$, the normalizing Möbius transformation being μ (so $\mu f \mu^{-1} = F_s$). Choose a path s_t, $0 \le t \le 1$, in U from a to s. The five points $0, \infty, 1, s_t, v(s_t)$ stay distinct, so by isotopy extension there is an ambient isotopy H_t of $\mathbb{P}^1$ with $H_0 = \mathrm{id}$ fixing $0, \infty, 1$ and with $H_t(a) = s_t$, $H_t(v_a) = v(s_t)$. Put $\gamma = \mu^{-1}(H_1(\gamma_a)) \subset \mathbb{P}^1 \setminus B$; equivalently, $H_1(\gamma_a)$ is the image of $H_1(C)$ under two twists about $H_1(\eta)$.*

The isotopy class of γ relative to B depends on the path s_t only through its homotopy class in U; Theorem 5.2 holds for every choice.

Theorem 5.2. *Let f, A, B be as in Section 3.2, and let γ be the curve of Construction 5.1. Then the Dehn twist T_γ preserves $\mathcal{D}_f$. If D_0 denotes the connected component of $\mathcal{D}_f$ containing the base point τ_0, then the components $T_\gamma^n(D_0)$, $n \in \mathbb{Z}$, are pairwise distinct.*

Sections 6–9 prove the theorem for the model $f = F_a$ and $\gamma = \gamma_a$; Section 10 deduces the general case.

6 A finite cover of the moduli space

Recall Y and M from (4.1); M is the moduli space of B_a, with $0, \infty, 1$ normalized and (y, z) the positions of the points corresponding to a and v_a. A point of $\mathcal{T}(B_a)$ determines, through (3.1), a quadratic rational map with marked source and target points; the normalized positions of the fourth source and target points give a map from $\mathcal{T}(B_a)$ to a surface $\mathcal{W}$ of marked maps, a finite cover of M. This is the parameter surface of Hironaka and Koch [HKa, §§4.1–4.2]; in general such spaces are Hurwitz spaces, finite coverings of the moduli space of the target [Ko, Thm. 2.6], [KPS, §3.1].

Consider a quadratic rational map with marked source points $0, \infty, 1, x$ and marked target points $0, \infty, 1, y, z$ realizing the portrait

$$0 \overset{2}{\longmapsto} \infty, \qquad \infty \longmapsto 1, \qquad 1 \longmapsto y, \qquad x \longmapsto 0, \tag{6.1}$$

where the 2 indicates a critical point, and having z as its other critical value. A degree-two map with a double pole at 0 and value 1 at ∞ is $(t - x)(t - r)/t^2$ with zeros x and r, and the condition $1 \mapsto y$ determines r. Hence the map is unique:

$$F_{x,y}(t) = \frac{(t - x)(t - r)}{t^2}, \qquad r = \frac{x + y - 1}{x - 1}. \tag{6.2}$$

Its critical points are 0 and $2xr/(x+r)$, and the critical value at the latter is $-(x-r)^2/(4xr)$, which in terms of x, y reads

$$Z(x,y) = \frac{((x-1)^2-y)^2}{4x(x+y-1)(1-x)}. \tag{6.3}$$

The marked points must be distinct and the map must have degree two ($r \neq 0$), so the parameter surface and its projection to M are

$$\mathcal{W} = \{(x,y) \in Y^2 : x+y \neq 1,\ Z(x,y) \notin \{0,1,y\}\}, \tag{6.4}$$

$$p\colon \mathcal{W} \longrightarrow M, \qquad p(x,y) = (y, Z(x,y)). \tag{6.5}$$

The maps F_s of the family (3.3) are the marked maps whose fourth source and target points agree: indeed $F_{s,s} = F_s$, the two expressions being quadratic maps with the same poles, zeros and value at ∞, and $Z(s,s) = v(s)$. The conditions $Z(s,s) \neq 0, 1, s$ are precisely the three quadratic factors in (5.2), so the *diagonal* of $\mathcal{W}$ is

$$\mathcal{V} = \mathcal{W} \cap \{x = y\} = \{(s,s) : s \in U\}, \tag{6.6}$$

a connected Riemann surface, since U is connected.

Lemma 6.1. *1. The map p is an unramified covering of degree four, and $\mathcal{W}$ is connected.*

2. For the model (5.3)*, with $\pi_A\colon \mathcal{T}(A_a) \to Y$ and $\pi_B\colon \mathcal{T}(B_a) \to M$ the projections of Section 4, the map*

$$q = (\pi_A \circ F_a^*,\ \pi_A \circ i_*)\colon \mathcal{T}(B_a) \longrightarrow \mathcal{W}$$

is well defined and satisfies $p \circ q = \pi_B$ and $q(\tau_0) = (a,a)$, and it is the universal covering of $\mathcal{W}$. Thus π_B factors through $\mathcal{W}$, and, with $\mathrm{pr}_x, \mathrm{pr}_y$ the two coordinate projections of $\mathcal{W}$, by construction

$$\pi_A \circ F_a^* = \mathrm{pr}_x \circ q, \qquad \pi_A \circ i_* = \mathrm{pr}_y \circ q. \tag{6.7}$$

> **Reviewer 1**
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> A commutative diagram would be effective

Proof. (1) Fix $(y,z) \in M$. The equation $Z(x,y) = z$ is

$$((x-1)^2 - y)^2 + 4zx(x-1)(x+y-1) = 0,$$

a monic polynomial equation of degree four in x. Its values at $x = 0, 1, 1-y$ are $(1-y)^2$, y^2 and $y^2(y-1)^2$, all nonzero, so every root satisfies $x \in Y$ and $x+y \neq 1$; since $z \notin \{0,1,y\}$, every root gives a point of $\mathcal{W}$. Thus $p^{-1}(y,z)$ consists of the roots of this equation. The roots are distinct because of the factorizations

$$\begin{aligned}
Z - 1 &= -\frac{(x^2+y-1)^2}{4x(x-1)(x+y-1)},\\
Z - y &= -\frac{(x^2+2xy-2x-y+1)^2}{4x(x-1)(x+y-1)},\\
\frac{\partial Z}{\partial x} &= -\frac{(x^2+y-1)\,((x-1)^2-y)\,(x^2+2xy-2x-y+1)}{4x^2(x-1)^2(x+y-1)^2},
\end{aligned}$$

which show that $\partial Z/\partial x$ vanishes only where $Z = 1$, $Z = 0$ or $Z = y$, and these values of z are excluded in M. So p has exactly four preimages over each point, $\partial Z/\partial x \neq 0$ on $\mathcal{W}$, and the implicit function theorem shows that p is a local homeomorphism. A local homeomorphism between Hausdorff spaces all of whose fibres have the same finite cardinality is a covering (intersect the images of disjoint neighbourhoods of the four preimages of a point, and count), so p is a covering of degree four. Finally $\mathcal{W}$ is the complement in $\mathbb{C}^2$ of the finitely many algebraic curves listed in (6.4), and such a complement is path connected: join two points by two complex lines through a generic auxiliary point, each meeting the curves in finitely many points, which a path in the line can avoid.

(2) Let $\tau = [\phi] \in \mathcal{T}(B_a)$, with ϕ normalized to fix $0, \infty, 1$, and let $F \circ \psi = \phi \circ F_a$ be a pullback diagram (3.1) with ψ also normalized to fix $0, \infty, 1$ (possible since ψ is unique up to a Möbius transformation). Put $x = \psi(a)$, $y = \phi(a)$, $z = \phi(v_a)$. Applying the diagram to the points of A_a and to the critical points $0, -6$ of F_a shows that F has the portrait (6.1) with marked source points $\{0, \infty, 1, x\}$, marked target points $\{0, \infty, 1, y, z\}$, and other critical value $F(\psi(-6)) = z$. Hence $F = F_{x,y}$ and $Z(x, y) = z$ by the uniqueness statement above; $x, y \in Y$ and $z \notin \{0, 1, y\}$ because ψ, ϕ are injective, and $x + y \neq 1$ because F has degree two. Therefore $(x, y) = (\pi_A F_a^* \tau, \pi_A i_* \tau) \in \mathcal{W}$ and $p(x, y) = (y, z) = \pi_B(\tau)$. For τ_0 we may take $\psi = \phi = \mathrm{id}$, so $q(\tau_0) = (a, a)$; and q is continuous because F_a^*, i_*, π_A are.

It remains to see that q is a covering. Let $N \subsetneq M$ be an open ball evenly covered by both p and π_B, with $p^{-1}(N) = \bigsqcup_j N_j$ and $\pi_B^{-1}(N) = \bigsqcup_k \widetilde{N}_k$. The connected set $q(\widetilde{N}_k)$ lies in a single N_j, where $q|_{\widetilde{N}_k} = (p|_{N_j})^{-1} \circ \pi_B|_{\widetilde{N}_k}$ is a homeomorphism onto N_j. Hence each N_j is evenly covered by q (possibly with empty preimage), so $q(\mathcal{T}(B_a))$ is open and closed in the connected space $\mathcal{W}$, and q is a surjective covering; as $\mathcal{T}(B_a)$ is simply connected, it is the universal covering. □

7 When a loop preserves a component

We now set up the covering-space criterion of Hironaka and Koch [HK, §§2.8–2.11] in the coordinates of Lemma 6.1. Let $b = (a, a) = q(\tau_0)$. A loop g in $\mathcal{W}$ based at b determines, via the loop $p \circ g$ in M and (4.2), a deck transformation δ of π_B, which sends τ_0 to the endpoint τ_1 of the lift of g to $\mathcal{T}(B_a)$ starting at τ_0. It is also a deck transformation of q: $q \circ \delta$ and q are both lifts of π_B through p, and they agree at τ_0 because $q(\tau_1) = b$. We denote δ by the same letter g and write $g(\tau)$, as in Section 4. The two *coordinate loops* of g are the loops $\mathrm{pr}_x \circ g$ and $\mathrm{pr}_y \circ g$ in Y based at a.

Lemma 7.1. *Let $E = \mathrm{im}\big(\pi_1(\mathcal{V}, b) \to \pi_1(\mathcal{W}, b)\big)$ be the set of classes of loops in $\mathcal{W}$ that are homotopic in $\mathcal{W}$ to a loop in the diagonal $\mathcal{V}$. Let $g \in \pi_1(\mathcal{W}, b)$, acting on $\mathcal{T}(B_a)$ as above, and let D_0 be the connected component of $\mathcal{D}_{F_a}$ containing τ_0.*

1. *If the two coordinate loops of g have equal classes in $\pi_1(Y, a)$, then $g(\mathcal{D}_{F_a}) = \mathcal{D}_{F_a}$.*
2. *$g(D_0) = D_0$ if and only if $g \in E$.*

Consequently, if the coordinate loops of g are homotopic and $g^n \notin E$ for every $n \neq 0$, then g preserves $\mathcal{D}_{F_a}$ and the components $g^n(D_0)$, $n \in \mathbb{Z}$, are pairwise distinct.

Proof. (1) By (6.7), F_a^* and i_* are lifts through the universal covering $\pi_A \colon \mathcal{T}(A_a) \to Y$ of the maps $\mathrm{pr}_x \circ q$ and $\mathrm{pr}_y \circ q$, and they take the same value at τ_0 because $\tau_0 \in \mathcal{D}_{F_a}$. Since g is a deck transformation of q, the map $F_a^* \circ g$ is another lift of $\mathrm{pr}_x \circ q$, so $F_a^* \circ g = \delta_x \circ F_a^*$ for a deck

transformation δ_x of π_A. To identify δ_x, lift g to a path $\tilde{g}$ from τ_0 to $g(\tau_0)$; then $F_a^* \circ \tilde{g}$ is the lift, starting at $F_a^*(\tau_0)$, of the loop $\mathrm{pr}_x \circ g$, so δ_x is the deck transformation of π_A determined by the class of $\mathrm{pr}_x \circ g$ relative to the base point $F_a^*(\tau_0)$. Likewise $i_* \circ g = \delta_y \circ i_*$ with δ_y determined by the class of $\mathrm{pr}_y \circ g$ relative to the same base point $i_*(\tau_0) = F_a^*(\tau_0)$. Equal classes give $\delta_x = \delta_y =: \delta$, and then for $\tau \in \mathcal{D}_{F_a}$,

$$F_a^*(g\tau) = \delta F_a^*(\tau) = \delta\, i_*(\tau) = i_*(g\tau),$$

so $g(\mathcal{D}_{F_a}) \subset \mathcal{D}_{F_a}$.

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> Since the action (4.2) is a right action, $g \mapsto$(deck transformation) is an anti-homomorphism $\pi_1(W, b) \to$ Deck. This is harmless here, because only powers of a single g are used, but it is worth restating at this point.

The coordinate loops of g^{-1} are the inverses of those of g, hence also homotopic, so $g^{-1}(\mathcal{D}_{F_a}) \subset \mathcal{D}_{F_a}$ as well.

(2) First, $\mathcal{D}_{F_a} \subset q^{-1}(\mathcal{V})$: if $F_a^*\tau = i_*\tau$, the two coordinates of $q(\tau)$ agree by (6.7). Next, on a connected component N of $q^{-1}(\mathcal{V})$, the maps $F_a^*|_N$ and $i_*|_N$ are lifts through π_A of one and the same map $\tau \mapsto \mathrm{pr}_x\, q(\tau) = \mathrm{pr}_y\, q(\tau)$, so by uniqueness of lifts they agree on all of N if they agree at one point. Hence $\mathcal{D}_{F_a}$ is a union of connected components of $q^{-1}(\mathcal{V})$; these are path connected, $q\colon q^{-1}(\mathcal{V}) \to \mathcal{V}$ being a covering of the connected manifold $\mathcal{V}$, and D_0 is the component through τ_0.

The deck transformation g permutes the components of $q^{-1}(\mathcal{V})$, so $g(D_0) = D_0$ if and only if there is a path in D_0 from τ_0 to $g(\tau_0)$. Given such a path, its image under q is a loop in $\mathcal{V}$ based at b, homotopic in $\mathcal{W}$ to g because both lift to paths in the simply connected space $\mathcal{T}(B_a)$ with the same endpoints; so $g \in E$. Conversely, if g is homotopic in $\mathcal{W}$ to a loop g' in $\mathcal{V}$, the lift of g' starting at τ_0 stays in $q^{-1}(\mathcal{V})$, hence in D_0, and ends at $g(\tau_0)$ by homotopy lifting; so $g(\tau_0) \in D_0$.

Finally, if $g^j(D_0) = g^k(D_0)$ with $j \neq k$, then $g^{j-k}(D_0) = D_0$, so $g^{j-k} \in E$ by (2), contrary to hypothesis. □

8 Loops in $\mathcal{W}$ representing the twist about γ_a

The goal of this section is a loop g in $\mathcal{W}$ with $P(p \circ g) = T_{\gamma_a}^{\pm 1}$ and trivial coordinate loops (Proposition 8.3). We work in the fibre $\mathcal{W}_a = \{x \in \mathbb{C} : (x, a) \in \mathcal{W}\}$ over $y = a = 3/5$, along which p fixes the four points of A_a and moves only v_a, so that mapping classes are point pushes of v_a. By (6.4) and the factorizations in the proof of Lemma 6.1, the points of $\mathbb{C}$ excluded from $\mathcal{W}_a$ are

$$0,\ 1,\ \tfrac{2}{5}, \qquad 1 \pm \sqrt{\tfrac{3}{5}}\ \ (Z = 0), \qquad \pm\sqrt{\tfrac{2}{5}}\ \ (Z = 1), \qquad \tfrac{2}{5} \pm \tfrac{i\sqrt{6}}{5}\ \ (Z = a); \tag{8.1}$$

here $2/5 = 1 - a$ is the pole of $Z(\cdot, a)$.

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> Incorrect that it is 'the' pole; $Z(, a)$ also has simple poles at 0 and 1, where the numerator is $(1-a)^2$ and a^2. 2/5 is the only pole not already excluded by $x \in Y$. Suggest 'the remaining pole'.

Their moduli are 0, 1, 0.4, 1.77…, 0.22…, and $\sqrt{2/5} = 0.632\ldots$ for the last four; none is $a = 0.6$, so $\ell(\theta) = (ae^{2\pi i\theta}, a)$, $0 \le \theta \le 1$, is a loop in $\mathcal{W}$ based at b. Its image $p \circ \ell$ in M moves v_a along the loop $\zeta(\theta) = Z(ae^{2\pi i\theta}, a)$ in $\mathbb{P}^1 \setminus A_a$ based at $v_a = Z(a, a)$.

Let m be a positive meridian about the pole $x = 2/5$ in $\mathcal{W}_a$: travel from a along the real interval toward $2/5$, go once counterclockwise around a small circle centred at $2/5$, and return along the interval.

Lemma 8.1. *The mapping class $P(p \circ m)$ is $T_C^{\pm 1}$, a single Dehn twist about the round circle C of (5.4).*

Proof. On $(2/5, 3/5]$ the factorizations give $\partial Z(x,a)/\partial x < 0$, because in the numerator of $\partial Z/\partial x$ the factors $x^2 - 2/5$ and $(x-1)^2 - 3/5$ are negative and $x^2 - \frac{4}{5}x + \frac{2}{5}$ is positive. Moreover $Z(a,a) = v_a$, so $Z(x,a) \geq v_a > 1$ on $(2/5, a]$ by monotonicity;

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> The inequality actually needed is $v_a = 242/240 > 241/240$, so that the approach path lies in the interior of Δ; > 1 is not the relevant bound. Worth stating, since the margin is only $1/240$ and it explains the choice of the radius of C. (The sign analysis of $\partial Z/\partial x$ on $(2/5, 3/5]$ is correct.)

the pole of $Z(\cdot, a)$ at $2/5$ is simple, and $Z(x,a) \to +\infty$ as $x \downarrow 2/5$. Thus $p \circ m$ moves v_a out along the positive real ray, once around ∞ (a small circle about a simple pole maps to a loop winding once about ∞), and back. For a small enough circle this motion takes place in the closed disk $\Delta = \{|z| \geq 241/240\} \cup \{\infty\}$ bounded by C, whose only marked points are v_a and ∞; in $\Delta \setminus \{\infty\}$, whose fundamental group is $\mathbb{Z}$, the loop is homotopic to a simple loop once around ∞, so (4.3) gives $P(p \circ m) = T_C^{\pm 1}$. □

The homotopy class of ζ, not merely its homology class, must be identified, since the isotopy class of γ_a depends on it.

Lemma 8.2. *The loop ζ is homotopic, as a loop based at v_a in $\mathbb{P}^1 \setminus A_a$, to two clockwise circuits around the segment $[a, 1]$ inside the disk bounded by η. Consequently $h := P(\zeta^{-1})$ satisfies $h(C) = T_\eta^2(C) = \gamma_a$, with the sign of Convention 4.1.*

Proof. Step 1: enlarging the circle. Put $R = \sqrt{1-a} = \sqrt{2/5} > a$. No excluded point in (8.1) has modulus in $[a, R)$, so for $a \leq r < R$ the loops $\zeta_r(\theta) = Z(re^{2\pi i\theta}, a)$ avoid $0, a, 1, \infty$ and depend continuously on r. Their base points $Z(r,a)$ move along the real path $\sigma_r = Z([a,r], a) \subset (1, v_a]$, since $Z(\cdot, a)$ is decreasing on $(2/5, R)$ (the sign analysis in the proof of Lemma 8.1 is valid there) with $Z(R,a) = 1$. Hence ζ is based-homotopic in $\mathbb{P}^1 \setminus A_a$ to $\sigma_r^{-1} \zeta_r \sigma_r$ for every $r \in [a, R)$.

Step 2: the circle $|x| = R$ maps onto $[a, 1]$, and nothing inside it does. The substitution $u = x + (1-a)/x$ (a Joukowski map) simplifies Z: writing $(x-1)^2 - a = x(u-2)$ and $(x + a - 1)(1-x) = x(2-a-u)$ gives

$$Z(x,a) = G(u), \qquad G(u) = \frac{(u-2)^2}{4(2-a-u)}. \tag{8.2}$$

For $x = Re^{i\vartheta}$ we have $u = 2R\cos\vartheta$, so on $|x| = R$ the variable u runs over $[-2R, 2R]$, where $2 - a - u \geq 2 - a - 2R > 0$ and

$$G(u) - 1 = \frac{u^2 - 4(1-a)}{4(2-a-u)}, \qquad G(u) - a = \frac{(u - 2(1-a))^2}{4(2-a-u)}.$$

Thus on $[-2R, 2R]$ the function G takes values in $[a, 1]$, equals 1 exactly at the endpoints, and equals a exactly at $u = 2(1-a)$; since $G'(u) = (u-2)(2-2a-u)/(4(2-a-u)^2)$, G decreases from 1 to a

and then increases back to 1. Hence for $z \in (a,1)$ the equation $G(u) = z$ has two distinct solutions $u \in (-2R, 2R)$, and for each of them the two roots of $x^2 - ux + (1-a) = 0$ are non-real conjugates with $|x|^2 = 1-a$, hence lie on $|x| = R$. This produces four distinct points of $|x| = R$ mapping to z, which exhaust the four preimages of (a, z) under p. For the endpoints the factorizations of $Z - 1$ and $Z - a$ show that all preimages of 1 and of a (namely $\pm R$ and $2/5 \pm i\sqrt{6}/5$, with multiplicity) also lie on $|x| = R$. So $Z(\{|x| = R\}, a) = [a, 1]$, and no x with $|x| < R$ maps into $[a, 1]$.

Step 3: the homotopy class. The segment $[a, 1]$ lies in the open disk bounded by η, so by Step 2 and continuity, for $r < R$ close to R the loop ζ_r lies in that disk and avoids $[a, 1]$; the base point path $\sigma_r \subset (1, v_a]$ also lies in the disk and avoids $[a, 1]$.

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> The continuity step uses that $Z(, a)$ is **uniformly** continuous on a compact annulus around $|x| = R$. This holds because its poles $0, 2/5$ and 1 are off that circle.

So $\sigma_r^{-1}\zeta_r\sigma_r$ is a loop in the η-disk minus the segment $[a, 1]$. This space is an annulus, with fundamental group $\mathbb{Z}$ generated by a counterclockwise circuit around $[a, 1]$, and the class of a loop is its winding number about any point of the segment, say about 1. Since the homotopy of Step 1 avoids 1, this winding number equals $\mathrm{wind}_1(\zeta)$, which we compute by the argument principle applied to

$$Z(x, a) - 1 = -\frac{(x^2 + a - 1)^2}{4x(x-1)(x+a-1)}$$

on the disk $|x| < a$: the numerator vanishes only at $\pm R$, outside the disk, while the denominator has the two simple zeros 0 and $1 - a = 2/5$ inside it. Hence $\mathrm{wind}_1(\zeta) = 0 - 2 = -2$, and ζ is based-homotopic to two clockwise circuits around $[a, 1]$ in the η-disk, as claimed.

Step 4: the point push. Choose a round circle κ enclosing $a, 1$ but not v_a and lying inside both C and η; for instance $\kappa = \{|z - \frac{4}{5}| = \frac{97}{480}\}$, for which $|a - \frac{4}{5}| = |1 - \frac{4}{5}| = \frac{96}{480} < \frac{97}{480} < \frac{100}{480} = |v_a - \frac{4}{5}|$, and the largest modulus on κ is $\frac{481}{480} < \frac{241}{240}$. The annulus between κ and η contains the single marked point v_a, and by Step 3, ζ^{-1} is homotopic to two counterclockwise circuits of v_a around this annulus. By Convention 4.1 one such circuit has point push $T_\eta T_\kappa^{-1}$, so, the twists about the disjoint curves κ and η commuting,

$$h = P(\zeta^{-1}) = T_\eta^2 T_\kappa^{-2}.$$

Since κ is disjoint from C, the twist T_κ fixes the isotopy class of C, and $h(C) = T_\eta^2(C) = \gamma_a$. □

Proposition 8.3. *Let*

$$g = \ell^{-1} m \ell \in \pi_1(\mathcal{W}, b), \tag{8.3}$$

the loop that traverses ℓ, then m, then ℓ backwards. Then $P(p \circ g) = T_{\gamma_a}^{\pm 1}$: the deck transformation g acts on $\mathcal{T}(B_a)$ as the Dehn twist about γ_a or its inverse. Both coordinate loops of g are null-homotopic in Y. In particular T_{γ_a} preserves $\mathcal{D}_{F_a}$.

Proof. Since $p \circ \ell$ moves only v_a, along ζ, we have $P(p \circ \ell) = P(\zeta) = h^{-1}$ with h as in Lemma 8.2. Multiplicativity of P (Section 4) and Lemma 8.1 give

$$P(p \circ g) = P(p \circ \ell^{-1})\, P(p \circ m)\, P(p \circ \ell) = h\, T_C^{\pm 1}\, h^{-1} = T_{h(C)}^{\pm 1} = T_{\gamma_a}^{\pm 1},$$

using (P2) and Lemma 8.2. The y-coordinate of g is constant. The x-coordinate of m is a small loop around $2/5$ together with a real approach path in $(2/5, 3/5]$; this loop lies in the disk $|x - \frac{1}{2}| < \frac{1}{2}$, which avoids 0 and 1, so it is null-homotopic in Y; hence the x-coordinate loop of g is homotopic to $(\mathrm{pr}_x \circ \ell)^{-1}(\mathrm{pr}_x \circ \ell)$, which is trivial. Lemma 7.1(1) now shows that g, hence T_{γ_a}, preserves $\mathcal{D}_{F_a}$. □

9 No nonzero power of the twist preserves the base component

By Lemma 7.1 and Proposition 8.3, it remains to show that $g^n \notin E$ for $n \neq 0$. Consider

$$\Phi\colon \mathcal{W} \longrightarrow Y \times \mathbb{C}^*, \qquad \Phi(x,y) = \big(t(x,y), L(x,y)\big) = \left(\frac{xy}{(1-x)(1-y)},\ (1-x)(1-y)\right). \tag{9.1}$$

It is well defined: $L \neq 0$ and $t \neq 0$ since $x, y \in Y$, and $t - 1 = (x+y-1)/((1-x)(1-y)) \neq 0$. In terms of the symmetric functions $e_1 = x+y$, $e_2 = xy$ one has $L = 1 - e_1 + e_2$ and $t = e_2/L$, so Φ is the symmetric quotient $(x,y) \mapsto (x+y, xy)$ of Hironaka–Koch [HK, §2.12], restricted to $\mathcal{W}$, up to this change of coordinates.

The constraint on diagonal loops. Enlarge the diagonal by filling in its six quadratic punctures: $\widehat{\mathcal{V}} = \{(s,s) : s \in \mathbb{C} \setminus \{0, 1, 1/2\}\} \supset \mathcal{V}$, on which (9.1) still makes sense, so that the covering (9.3) below covers all of Y. In the coordinate $w = s/(1-s)$, which maps $\mathbb{C} \setminus \{0, 1, 1/2\}$ bijectively onto $\mathbb{C} \setminus \{0, 1, -1\}$, the restriction of Φ is

$$t = w^2, \qquad L = (1+w)^{-2}, \qquad w \in \mathbb{C} \setminus \{0, 1, -1\}, \tag{9.2}$$

because $1 + w = 1/(1-s)$. The squaring map

$$\mathbb{C} \setminus \{0, 1, -1\} \longrightarrow Y, \qquad w \longmapsto w^2, \tag{9.3}$$

is a two-sheeted covering; our base points are $w_0 = a/(1-a) = 3/2$ and $t_0 = w_0^2 = 9/4$.

Lemma 9.1. *Let λ be a loop in $\mathcal{V}$ based at b. Then $w \circ \lambda$ is the lift of the loop $t \circ \lambda$ through* (9.3) *starting at w_0; it is closed, and*

$$\operatorname{wind}_0(L \circ \lambda) = -2 \operatorname{wind}_{-1}(w \circ \lambda). \tag{9.4}$$

The based homotopy class of $w \circ \lambda$ in $\mathbb{C} \setminus \{0, 1, -1\}$, hence the right-hand side of (9.4)*, depends only on the based homotopy class of $t \circ \lambda$ in Y.*

Proof. On $\mathcal{V} \subset \widehat{\mathcal{V}}$ formula (9.2) applies: $w \circ \lambda$ is a closed path in $\mathbb{C} \setminus \{0, 1, -1\}$ from w_0 with $(w \circ \lambda)^2 = t \circ \lambda$, hence the lift of $t \circ \lambda$ from w_0, and $L \circ \lambda = (1 + w \circ \lambda)^{-2}$ gives (9.4). The last assertion is homotopy lifting for the covering (9.3). □

The loops g^n violate the constraint. On the fibre $y = a$ the map (9.1) becomes

$$t = \frac{3x}{2(1-x)}, \qquad L = \frac{2}{5}(1-x). \tag{9.5}$$

Let $\alpha = t \circ \ell$ and $\beta = t \circ m$, loops in Y based at t_0.

Lemma 9.2. *For every integer n, the loop g^n satisfies*

$$\operatorname{wind}_0(L \circ g^n) = 0, \qquad \operatorname{wind}_{-1}\big(\widetilde{t \circ g^n}\big) = n, \tag{9.6}$$

where $\widetilde{t \circ g^n}$ is the lift of $t \circ g^n$ through (9.3) *starting at w_0, which is a closed loop.*

Proof. The loop ℓ. Along ℓ we have $x = ae^{2\pi i\theta}$ with $|x| = a < 1$, so $1 - x$ does not wind about 0 and (9.5) gives $\mathrm{wind}_0(\alpha) = 1$ and $\mathrm{wind}_0(L \circ \ell) = 0$. Since $\mathrm{wind}_0(\alpha)$ is odd, continuation of the square root along α, or along α^{-1}, interchanges the two sheets of (9.3): the lift $\tilde{\alpha}$ of α starting at w_0 ends at $-w_0$.

The loop m. The Möbius map $x \mapsto t$ in (9.5) sends $2/5$ to 1 and a to $9/4$, is increasing on the real interval $(2/5, a]$, which it maps onto $(1, 9/4]$, and, being conformal, sends a small counterclockwise circle about $2/5$ to a small counterclockwise loop about 1. Hence β is a positive meridian about 1 in Y, approached from t_0 along the real axis to the right of 1. Its lift starting at $w_0 = 3/2$ runs along $(1, 3/2]$ and around a small loop about $+1$, so it is a positive meridian about $+1$; the lift starting at $-w_0$ is its negative,

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> 'Its negative' is ambiguous next to loop inverses: say 'its image under $w \mapsto -w$'. Since this map is a rotation, it preserves orientation, which is what gives a positive meridian about -1.

a positive meridian about -1 (the map $w \mapsto -w$ preserves orientation). Also $\mathrm{wind}_0(L \circ m) = 0$, since $1 - x$ stays near the positive interval $[2/5, 3/5]$ along m.

The loop g. By additivity, $\mathrm{wind}_0(L \circ g) = -\mathrm{wind}_0(L \circ \ell) + \mathrm{wind}_0(L \circ m) + \mathrm{wind}_0(L \circ \ell) = 0$, and so $\mathrm{wind}_0(L \circ g^n) = 0$. The t-projection of g is $\alpha^{-1}\beta\alpha$: first α, then β, then α backwards. Its lift from w_0 consists of $\tilde{\alpha}$, ending on the other sheet at $-w_0$, then the lift of β from $-w_0$, a positive meridian about -1, then $\tilde{\alpha}$ reversed, returning to w_0. The contributions of $\tilde{\alpha}$ and its reverse to the change of argument of $w + 1$ cancel, so this closed lift has winding number 1 about -1, and the lift of $t \circ g^n = (\alpha^{-1}\beta\alpha)^n$ from w_0, the n-fold concatenation, has winding number n. □

Proposition 9.3. *For every $n \neq 0$, $g^n \notin E$. Consequently Theorem 5.2 holds for the model map F_a and the curve $\gamma_a = T_\eta^2(C)$.*

Proof. Suppose $g^n \in E$, so that g^n is homotopic in $\mathcal{W}$ to a loop λ in $\mathcal{V}$ based at b. Applying Φ, the loops $t \circ \lambda$ and $t \circ g^n$ are based-homotopic in Y, and $L \circ \lambda$ and $L \circ g^n$ are homotopic in $\mathbb{C}^*$. By Lemma 9.1, the lift $w \circ \lambda$ is based-homotopic to the lift of $t \circ g^n$ used in Lemma 9.2, so $\mathrm{wind}_{-1}(w \circ \lambda) = n$, while $\mathrm{wind}_0(L \circ \lambda) = \mathrm{wind}_0(L \circ g^n) = 0$. The constraint (9.4) then reads $0 = -2n$, so $n = 0$.

For the second assertion, g acts as $T_{\gamma_a}^{\pm 1}$ and has homotopic coordinate loops (Proposition 8.3), so by Lemma 7.1 the twist T_{γ_a} preserves $\mathcal{D}_{F_a}$ and the components $T_{\gamma_a}^n(D_0) = g^{\pm n}(D_0)$, $n \in \mathbb{Z}$, are pairwise distinct. □

Remark 9.4. The same computation applies to the other conjugate $g' = \ell m \ell^{-1}$ (its t-projection $\alpha\beta\alpha^{-1}$ also changes sheet before the meridian, as α^{-1} has odd winding number about 0, and again $\mathrm{wind}_0(L \circ g') = 0$), whose mapping class is $h^{-1}T_C^{\pm 1}h = T_{h^{-1}(C)}^{\pm 1}$ with $h^{-1}(C) = T_\eta^{-2}T_\kappa^2(C) = T_\eta^{-2}(C)$. Hence Theorem 5.2 holds for both curves $T_\eta^{\pm 2}(C)$.

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> To apply Lemma 7.1 to g', one also needs its coordinate loops to be homotopic. This holds exactly as in Prop. 8.3

10 Transport to an arbitrary map of the family

Let f, μ, F_s, s_t, H_t be as in Construction 5.1. We show that changing markings by $H = H_1$ carries $\mathcal{D}_{F_a}$ onto $\mathcal{D}_{F_s}$, D_0 to the base component of $\mathcal{D}_{F_s}$, and conjugates T_{γ_a} to $T_{H(\gamma_a)}$.

Lemma 10.1. *There is a continuous family of homeomorphisms K_t of $\mathbb{P}^1$, $0 \le t \le 1$, with $K_0 = \mathrm{id}$ and*

$$F_{s_t} \circ K_t = H_t \circ F_a \qquad (0 \le t \le 1). \tag{10.1}$$

It satisfies $K_t|_{A_a} = H_t|_{A_a}$ for all t; consequently $H_t^{-1}K_t$ is an isotopy from the identity through homeomorphisms fixing A_a pointwise, and K_1 is isotopic to H_1 relative to A_a.

Proof. Every F_s is Möbius-equivalent to $Q(z) = z^2$, with Möbius maps depending continuously on s: completing the square in $F_s(z) = 1 + b_s/z + c_s/z^2$ and recalling $v(s) = 1 - b_s^2/(4c_s)$ gives $F_s = v(s) + c_s\, J_s^2$, that is,

$$\nu_s \circ F_s = Q \circ J_s, \qquad J_s(z) = \frac{1}{z} + \frac{b_s}{2c_s}, \qquad \nu_s(w) = \frac{w - v(s)}{c_s},$$

with $c_s \neq 0$ for $s \in U$. The isotopy $\widehat{H}_t = \nu_{s_t} \circ H_t \circ \nu_a^{-1}$ starts at the identity and fixes 0 and ∞, because $H_t(v_a) = v(s_t)$ and $H_t(\infty) = \infty$. By the homotopy lifting property of the covering $Q\colon \mathbb{C}^* \to \mathbb{C}^*$, the homotopy $(z,t) \mapsto \widehat{H}_t(Q(z))$ lifts uniquely to a continuous $\rho\colon \mathbb{C}^* \times [0,1] \to \mathbb{C}^*$ with $\rho_0 = \mathrm{id}$ and $Q \circ \rho_t = \widehat{H}_t \circ Q$; lifting $\widehat{H}_t^{-1} \circ Q$ in the same way gives ρ_t' with $\rho_t'\rho_t = \mathrm{id} = \rho_t\rho_t'$ (both products lift the constant homotopy Q and start at id), so each ρ_t is a homeomorphism of $\mathbb{C}^*$, and ρ extends continuously to $\mathbb{P}^1 \times [0,1]$ by $\rho_t(0) = 0$, $\rho_t(\infty) = \infty$, since $|\rho_t(z)|^2 = |\widehat{H}_t(z^2)|$ and $\widehat{H}$ is continuous on the compact space $\mathbb{P}^1 \times [0,1]$. Then $K_t = J_{s_t}^{-1} \circ \rho_t \circ J_a$ is a continuous family of homeomorphisms with $K_0 = \mathrm{id}$ and $F_{s_t}K_t = \nu_{s_t}^{-1} Q \rho_t J_a = \nu_{s_t}^{-1}\widehat{H}_t Q J_a = H_t \nu_a^{-1} Q J_a = H_t F_a$, which is (10.1).

For the marked points, (10.1) gives $F_{s_t}(K_t(w)) = H_t(F_a(w))$ for $w \in A_a$. Thus $K_t(0)$ is the critical point of F_{s_t} over ∞, namely 0; $K_t(\infty)$ lies in $F_{s_t}^{-1}(1) = \{\infty, -c_{s_t}/b_{s_t}\}$, $K_t(1)$ in $F_{s_t}^{-1}(s_t)$, and $K_t(a)$ in $F_{s_t}^{-1}(0) \ni s_t$. For $s_t \in U$ each of the latter three fibres (over the non-critical values 1, s_t, 0) consists of two distinct points depending continuously on t: $b_{s_t} \neq 0$ since $s_t^2 + s_t - 1 \neq 0$, and s_t, 0 are not critical values of F_{s_t} since $v(s_t) \neq s_t, 0$.

> **Reviewer 1**
>
> Clearer to say explicitly: $\infty \in F^{-1}(1)$, $1 \in F^{-1}(s_t)$ and $s_t \in F^{-1}(0)$ for every t. Since each of these fibres has two distinct points, the continuous maps $t \mapsto K_t(\infty), K_t(1), K_t(a)$ cannot switch branches.

By continuity from $t = 0$, $K_t(0) = 0$, $K_t(\infty) = \infty$, $K_t(1) = 1$ and $K_t(a) = s_t$, which are the values of H_t on A_a. The remaining assertions follow. □

Proof of Theorem 5.2. Write $H = H_1$, $K = K_1$. Since H maps B_a onto B_s (notation of Section 5), changing markings by H is a biholomorphic isomorphism

$$\Psi\colon \mathcal{T}(B_a) \longrightarrow \mathcal{T}(B_s), \qquad [\phi] \longmapsto [\phi \circ H^{-1}],$$

and likewise $[\psi] \mapsto [\psi \circ H^{-1}]$ identifies $\mathcal{T}(A_a)$ with $\mathcal{T}(A_s)$, $A_s = \{0, \infty, 1, s\} = H(A_a) = K(A_a)$.

Ψ *carries $\mathcal{D}_{F_a}$ onto $\mathcal{D}_{F_s}$.* Let ϕ be a marking with pullback diagram $\widetilde{F} \circ \psi = \phi \circ F_a$ for F_a. Using $F_a \circ K^{-1} = H^{-1} \circ F_s$ from (10.1),

$$\widetilde{F} \circ (\psi K^{-1}) = \phi \circ F_a \circ K^{-1} = (\phi H^{-1}) \circ F_s,$$

so ψK^{-1} is a source marking for the target marking ϕH^{-1} in a pullback diagram for F_s: $F_s^*\Psi[\phi] = [\psi K^{-1}]$, while $i_*\Psi[\phi] = [\phi H^{-1}]$. Now $[\psi K^{-1}] = [\phi H^{-1}]$ in $\mathcal{T}(A_s)$ if and only if $[\psi K^{-1}H] = [\phi]$ in $\mathcal{T}(A_a)$, and $K^{-1}H$ is isotopic to the identity relative to A_a by Lemma 10.1, so this holds if and only if $[\psi] = [\phi]$ in $\mathcal{T}(A_a)$, that is, $[\phi] \in \mathcal{D}_{F_a}$. Hence $\Psi(\mathcal{D}_{F_a}) = \mathcal{D}_{F_s}$.

Ψ *conjugates twists.* For a mapping class $k \in \mathrm{PMod}(B_a)$, $\Psi([\phi] \cdot k) = [\phi k H^{-1}] = [\phi H^{-1}] \cdot (HkH^{-1})$, so Ψ intertwines the action of T_{γ_a} with that of $HT_{\gamma_a}H^{-1} = T_{H(\gamma_a)}$, by the naturality of twists (P2), which holds verbatim for the homeomorphism $H\colon (\mathbb{P}^1, B_a) \to (\mathbb{P}^1, B_s)$.

> **Reviewer 1**
>
> State explicitly that $T_H(\gamma_a)(D_{F_s}) = \Psi(T_{\gamma_a}(D_{F_a})) = \Psi(D_{F_a}) = D_{F_s}$. The proof currently records only the distinctness of the components, while Theorem 5.2 also asserts invariance.

Ψ *carries D_0 to the base component.* The path of markings $[H_t]$ in $\mathcal{T}(B_a)$ lies in $\mathcal{D}_{F_a}$: by (10.1), K_t is a source marking for the target marking H_t, and $[K_t] = [H_t]$ in $\mathcal{T}(A_a)$ because $H_t^{-1}K_t$ is isotopic to the identity relative to A_a (Lemma 10.1). The path starts at τ_0, so its endpoint $[H]$ lies in D_0, and $\Psi[H] = [\mathrm{id}]$ is the base point of $\mathcal{T}(B_s)$. Hence $\Psi(D_0)$ is the base component of $\mathcal{D}_{F_s}$, and the components $T^n_{H(\gamma_a)}(\Psi(D_0)) = \Psi(T^n_{\gamma_a}(D_0))$, $n \in \mathbb{Z}$, are pairwise distinct by Proposition 9.3. Finally, conjugating by μ^{-1} identifies $(\mathbb{P}^1, B_s, F_s)$ with $(\mathbb{P}^1, B, f)$ and $H(\gamma_a)$ with γ, which proves the theorem for f. □

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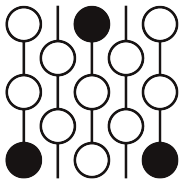
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Review of Problem 7 – ProofCouncil

First Proof Third Batch
Reviewer 2
October 8–9, 2026



1 Review

The article follows a seemingly sound strategy for producing the desired explicit description of a simple closed curve γ, such that the Dehn twist around γ has an infinite orbit on the components of the deformation space of quadratic maps with a critical 4-cycle.

Correctness. I believe that the solution is mostly correct; it is impossible to verify all the computational details within the time frame.
Score: 0.5

Exposition. The argument is possible to follow in principle, the exposition is, however, rather convoluted. Some of the computations and coordinate changes feel unnecessary, the notational choices make the article difficult to parse.
Score: -1

Attributions. The attributions are adequate. For some reason, there are references to two versions of the same article.
Score: 0

2 Submission with inline comments

Abstract

Let f be a quadratic rational map one of whose critical points lies in a cycle A of length four and whose other critical value v lies outside A, and let $B = A \cup \{v\}$. Epstein's deformation space $\mathcal{D}_f \subset \mathcal{T}(\mathbb{S}^2, B)$ is known to be disconnected (Hironaka–Koch). We write down an explicit simple closed curve γ on $(\mathbb{S}^2, B)$, for a normalized model map the image of a round circle under two Dehn twists about a second round circle, transported to a general f along a path in parameter space, and prove that the Dehn twist T_γ itself preserves $\mathcal{D}_f$ and that the translates $T_\gamma^n(D_0)$, $n \in \mathbb{Z}$, of the base component D_0 are pairwise distinct components, using the covering-space framework of Hironaka–Koch and a winding-number obstruction.

3 Introduction

3.1 Deformation spaces

Throughout, $\mathbb{P}^1$ is the Riemann sphere, which we also regard as the oriented topological sphere $\mathbb{S}^2$. For a finite set $B \subset \mathbb{P}^1$ with $|B| \geq 3$, the *Teichmüller space* $\mathcal{T}(B) = \mathcal{T}(\mathbb{S}^2, B)$ is the set of orientation-preserving homeomorphisms $\phi\colon \mathbb{P}^1 \to \mathbb{P}^1$ (*markings*), where $\phi \sim \phi'$ when $\phi' = \mu \circ \phi \circ h$ for a Möbius transformation μ and a homeomorphism h isotopic to the identity relative to B. Pulling back the standard complex structure by ϕ identifies $\mathcal{T}(B)$ with the complex structures on $(\mathbb{S}^2, B)$ up to isotopy relative to B; the class $\tau_0 = [\mathrm{id}]$ is the *base point*.

Let $f\colon \mathbb{P}^1 \to \mathbb{P}^1$ be an orientation-preserving branched cover of degree $d \geq 2$ with set of critical values $V(f)$, and let $A, B \subset \mathbb{P}^1$ be finite sets with $|A|, |B| \geq 3$ and $f(A) \cup V(f) \subset B$. Pulling back the complex structure of a marking ϕ under f gives a complex structure on $(\mathbb{S}^2, A)$; concretely, there are a homeomorphism ψ, unique up to post-composition with a Möbius transformation, and a rational map F with

$$F \circ \psi = \phi \circ f, \tag{3.1}$$

and $f^*[\phi] := [\psi] \in \mathcal{T}(A)$. This is the Thurston pullback map $f^*\colon \mathcal{T}(B) \to \mathcal{T}(A)$; it is holomorphic [BCT, §1.3, Prop. 1.4]. We call ϕ the *target marking* and ψ the *source marking* of the diagram (3.1). If $A \subset B$, forgetting the points of $B \setminus A$ gives the holomorphic *forgetful map* $i_*\colon \mathcal{T}(B) \to \mathcal{T}(A)$, $[\phi] \mapsto [\phi]$. When $f(A) \cup V(f) \cup A \subset B$, both maps are defined, and the *deformation space* of f is their equalizer

$$\mathcal{D}_f = \{\tau \in \mathcal{T}(B) : f^*(\tau) = i_*(\tau)\}.$$

This notion is due to Epstein [Ep]; see [BCT, §1.4] for an exposition. When f is rational, as in the problem considered here, we may take $\psi = \mathrm{id}$, $F = f$ in (3.1) for $\phi = \mathrm{id}$, so $\tau_0 \in \mathcal{D}_f$. (By Epstein's theorem [BCT, Thm. 1.5], under the standing hypotheses above, if f is not a topological $(2,2,2,2)$-map in the sense of [BCT, §1.4], then $\mathcal{D}_f$, whenever nonempty, is a smooth complex submanifold of $\mathcal{T}(B)$ of dimension $|B \setminus A|$. For the rational family considered here $\tau_0 \in \mathcal{D}_f$, and the periodic critical point excludes the exceptional type.)

The *pure mapping class group* $\mathrm{PMod}(B)$, the group of isotopy classes relative to B of orientation-preserving homeomorphisms of $\mathbb{P}^1$ fixing B pointwise, acts on $\mathcal{T}(B)$ on the right by

$$[\phi] \cdot h = [\phi \circ h]; \tag{3.2}$$

for a subset $D \subset \mathcal{T}(B)$ we write $h(D) = \{\tau \cdot h : \tau \in D\}$. A mapping class that preserves $\mathcal{D}_f$ permutes its connected components. For a simple closed curve γ in $\mathbb{P}^1 \setminus B$ we write $T_\gamma \in \mathrm{PMod}(B)$

for the Dehn twist about γ. The question addressed here asks for an explicit γ such that some power of T_γ preserves $\mathcal{D}_f$ and has an infinite orbit on the components.

3.2 The family and the result

Let f be a quadratic rational map with critical points c_1, c_2 such that c_1 lies in a cycle of length 4 and the critical value $v = f(c_2)$ is not in this cycle, and put

$$A = \{c_1, f(c_1), f^{\circ 2}(c_1), f^{\circ 3}(c_1)\}, \qquad B = A \cup \{v\}.$$

Then $f(A) = A$ and $V(f) = \{f(c_1), v\} \subset B$, so $\mathcal{D}_f \subset \mathcal{T}(B)$ is defined; $\mathcal{T}(B)$ has dimension 2. Conjugating f by the Möbius transformation that sends $c_1, f(c_1), f^{\circ 2}(c_1)$ to $0, \infty, 1$ normalizes the cycle to $0 \mapsto \infty \mapsto 1 \mapsto s \mapsto 0$, and Section 5 shows that this forces f to become

$$F_s(z) = 1 + \frac{-s^2 - s + 1}{(s-1)z} + \frac{s(2s-1)}{(s-1)z^2}, \qquad s \in U, \tag{3.3}$$

for an explicit Zariski-open set $U \subset \mathbb{C}$ given in (5.2).

> **Reviewer 2**
>
> An explanation of what is meant here by "our model" would be helpful

Our model is F_a with $a = 3/5$, whose marked set is $B_a = \{0, \infty, 1, a, v_a\}$, where $v_a = 121/120$ is its second critical value (computed in (5.1)).

The answer (Theorem 5.2) is as follows. Let C be the round circle $|z| = 241/240$, which separates $\{\infty, v_a\}$ from $\{0, 1, a\}$, and let η be the round circle $|z - 4/5| = 1/4$, which encloses exactly $a, 1, v_a$. Put $\gamma_a = T_\eta^2(C)$, the image of C under two Dehn twists about η (with the sign of Convention 4.1; either sign works, Remark 9.4). For a general f of the family, let γ be obtained from γ_a by following any path in U from a to the parameter of f and dragging the marked points along. Then T_γ preserves $\mathcal{D}_f$, and the translates $T_\gamma^n(D_0)$, $n \in \mathbb{Z}$, of the component $D_0 \ni \tau_0$ are pairwise distinct components; so the power of the twist can be taken to be 1. (A path is needed: γ_a and C separate B_a in the same way, so separation data do not determine the isotopy class of γ.)

> **Reviewer 2**
>
> The last sentence is a bit odd.

3.3 Relation to earlier work

Hironaka and Koch [HK] proved that for this family $\mathcal{D}_f$ has infinitely many connected components ([HK, Thm. 1.4]), introducing the tools we use: the surface of marked quadratic maps with its degree-four covering of moduli space and its diagonal ([HKa, §§4.1–4.2], [HK, §3]), the description of $\mathcal{D}_f$ through mapping classes that lift under f and through equalizers ([HK, §§2.8–2.11]), and a symmetric quotient of the parameter surface used as an obstruction

> **Reviewer 2**
>
> An obstruction to what? This is unclear

([HK, §2.12]); they also exhibit a mapping class preserving $\mathcal{D}_f$ with an infinite orbit on components ([HK, Prop. 3.2]). Here [HKa] is the first arXiv version of their paper and [HK] the revised author version, whose numbering we follow. Firsova, Kahn and Selinger [FKS] treat the same family

([FKS, §6]; numbering of arXiv v4), study twists along multicurves adapted to the pullback ("equalizing multicurves"), and prove that these deformation spaces are not contractible ([FKS, Lemma 8.4]). The present note adds the identification of a single, explicitly drawn Dehn twist with an infinite component orbit, which we have not found in [HK] or [FKS], with an elementary verification; our obstruction map Φ (Section 9) is a form of the symmetric quotient of [HK].

3.4 Strategy and outline

The proof has five steps. (1) Section 6 factors the universal cover $\mathcal{T}(B_a) \to M$ of the moduli space M of five marked points through a finite cover $\mathcal{W} \to M$, the surface of quadratic maps with the prescribed portrait of marked orbits; the pullback and forgetful maps become the coordinate projections of $\mathcal{W}$, and $\mathcal{D}_{F_a}$ lies over the diagonal $\mathcal{V} \subset \mathcal{W}$. (2) Section 7: a loop g in $\mathcal{W}$ acts on $\mathcal{T}(B_a)$ as a mapping class, which preserves $\mathcal{D}_{F_a}$ if the two coordinate loops of g are homotopic, and preserves the base component D_0 if and only if g is homotopic in $\mathcal{W}$ to a loop in $\mathcal{V}$. (3) Section 8 exhibits a loop $g = \ell^{-1} m \ell$ in $\mathcal{W}$ acting as $T_{\gamma_a}^{\pm 1}$: m is a meridian acting as the twist about C, and the point push along ℓ is identified exactly by a Joukowski substitution. (4) Section 9 shows that no power g^n, $n \neq 0$, is homotopic to a loop in $\mathcal{V}$, using a map $\Phi = (t, L) \colon \mathcal{W} \to (\mathbb{C} \setminus \{0,1\}) \times \mathbb{C}^*$ whose first coordinate becomes, on an enlarged diagonal, the two-sheeted covering $w \mapsto w^2$, while its second coordinate imposes a linear relation between two winding numbers that the loops g^n violate. (5) Section 10 transports the conclusion to every map of the family by lifting an isotopy of marked points through the branched covers.

4 Conventions on loops, point pushing and twists

For B as above, the *moduli space* $\mathcal{M}(B)$ is the set of injective maps $B \to \mathbb{P}^1$ modulo post-composition with Möbius transformations. Choosing three points of B and normalizing their images to $0, \infty, 1$ identifies $\mathcal{M}(B)$ with an open subset of $\mathbb{C}^{|B|-3}$. The projection $\pi_B \colon \mathcal{T}(B) \to \mathcal{M}(B)$, $[\phi] \mapsto [\phi|_B]$, is a holomorphic universal covering whose deck group is $\mathrm{PMod}(B)$ acting by (3.2); see [KPS, §2.1].

Two moduli spaces occur. For four marked points labelled $0, \infty, 1, x$, and for five points labelled $0, \infty, 1, y, z$, we use respectively

$$Y = \mathbb{C} \setminus \{0,1\}, \qquad M = \{(y,z) \in Y^2 : y \neq z\}. \tag{4.1}$$

Thus for $A \supset \{0, \infty, 1\}$ with $|A| = 4$, $\pi_A \colon \mathcal{T}(A) \to Y$ records the fourth point under the marking normalized to fix $0, \infty, 1$, and similarly $\pi_B \colon \mathcal{T}(B) \to M$ when $|B| = 5$.

Products of based loops are read from right to left: $\alpha\beta$ traverses β first and then α, matching composition of homeomorphisms. For a closed loop ω in $\mathbb{C} \setminus \{c\}$ we write $\mathrm{wind}_c(\omega) \in \mathbb{Z}$ for its winding number about c; for a non-closed path, $\mathrm{wind}_c(\omega)$ denotes the total change of argument of $\omega - c$ divided by 2π, which is additive under concatenation and changes sign under reversal.

Let δ be a loop in $\mathcal{M}(B)$ based at $\pi_B(\tau_0)$, that is, a motion of the marked points. By isotopy extension there is an ambient isotopy h_t of $\mathbb{P}^1$ with $h_0 = \mathrm{id}$ such that $h_t|_B$ follows $\delta(t)$. The mapping class $P(\delta) = [h_1] \in \mathrm{PMod}(B)$ depends only on the homotopy class of δ, and $[h_t]$ is the lift of δ to $\mathcal{T}(B)$ starting at τ_0, with endpoint $\tau_0 \cdot P(\delta)$. Consequently the deck transformation associated with δ, the one sending τ_0 to the endpoint of this lift, is

$$[\phi] \longmapsto [\phi \circ P(\delta)]. \tag{4.2}$$

Point pushing respects products, $P(\alpha\beta) = P(\alpha)P(\beta)$, since the isotopy following $\alpha\beta$ may be taken to be h_t^β followed by $h_t^\alpha \circ h_1^\beta$. Because (4.2) is a right action, deck transformations of products compose in the opposite order; we only use deck transformations of powers of a single loop. We write $g(\tau)$ for the deck transformation (4.2) of a loop g applied to τ.

When only one marked point v moves, δ is a loop ζ in $\mathbb{P}^1 \setminus (B \setminus \{v\})$ based at v, and $P(\zeta)$ is the *point-pushing map* $\mathrm{Push}(\zeta)$ of the Birman exact sequence

> **Reviewer 2**
>
> It would have probably been better to us Push instead of P throughout.

[FM, §4.2, Thm. 4.6]. We use two standard facts. (P1) If ζ is a simple loop, then $\mathrm{Push}(\zeta) = T_{\zeta_1} T_{\zeta_2}^{-1}$, where ζ_1, ζ_2 are the two boundary curves of a regular neighbourhood of ζ, the order being fixed by the orientation of ζ and the sign convention for twists [FM, Fact 4.7]. (P2) For every mapping class h and simple closed curve c, $hT_c h^{-1} = T_{h(c)}$ [FM, Fact 3.7].

Two consequences of (P1) are used. First, if $\Delta \subset \mathbb{P}^1$ is a closed disk with $\partial\Delta \cap B = \emptyset$ and $\Delta \cap B = \{v, w\}$, and ζ is a simple loop of v in $\Delta \setminus \{w\}$ going once around w, then one boundary curve of its regular neighbourhood bounds a disk containing only w (trivial twist) and the other is isotopic to $\partial\Delta$, so

$$\mathrm{Push}(\zeta) = T_{\partial\Delta}^{\pm 1}. \tag{4.3}$$

Second, if κ, η are disjoint simple closed curves bounding an annulus meeting B only in v, with κ inside η in a given disk picture,

> **Reviewer 2**
>
> "in a given disk picture" is unclear

and ζ is a simple loop of v once counterclockwise around the core of the annulus, then $\mathrm{Push}(\zeta) = T_\eta^\epsilon T_\kappa^{-\epsilon}$ with $\epsilon = \pm 1$ depending only on the sign convention for Dehn twists.

Convention 4.1. We fix the sign convention for Dehn twists so that $\epsilon = +1$, that is, $\mathrm{Push}(\zeta) = T_\eta T_\kappa^{-1}$ for such a counterclockwise circuit. With the opposite convention the boxed formula (5.5) of Section 5 reads $T_\eta^{-2}(C)$; both curves have the stated property (Remark 9.4).

5 The model map and the curve

Let f be as in Section 3.2 and conjugate it so that its critical cycle is $0 \mapsto \infty \mapsto 1 \mapsto s \mapsto 0$ with 0 the critical point. A quadratic rational map with a critical point at 0 mapping to ∞ has a double pole at 0 and no other pole, so it is $1 + b_s/z + c_s/z^2$ after using $F(\infty) = 1$. Solving the linear equations $F(1) = s$ and $F(s) = 0$ for b_s, c_s gives $b_s = (-s^2 - s + 1)/(s-1)$ and $c_s = s(2s-1)/(s-1)$, which is (3.3). The other critical point of F_s is $-2c_s/b_s$, and its critical value is $1 - b_s^2/(4c_s)$, which simplifies to

$$v(s) = -\frac{(s^2 - 3s + 1)^2}{4s(s-1)(2s-1)}. \tag{5.1}$$

The map has degree two and the four cycle points are distinct exactly when $s \neq 0, 1, 1/2$ (for $s = 1/2$ one has $c_s = 0$). The identities

$$v(s) - 1 = -\frac{(s^2 + s - 1)^2}{4s(s-1)(2s-1)}, \qquad v(s) - s = -\frac{(3s^2 - 3s + 1)^2}{4s(s-1)(2s-1)}$$

show that $v(s) \notin \{0, \infty, 1, s\}$, that is, $v(s)$ lies outside the cycle, exactly when the three quadratic factors below are nonzero. So the maps in the problem are, up to Möbius conjugacy, precisely the F_s with

$$s \in U = \left\{ s \in \mathbb{C} : s(s-1)(2s-1)(s^2-3s+1)(s^2+s-1)(3s^2-3s+1) \neq 0 \right\}. \tag{5.2}$$

The set U, a cofinite subset of $\mathbb{C}$, is connected. We write $A_s = \{0, \infty, 1, s\}$ and $B_s = A_s \cup \{v(s)\}$ for the marked sets of F_s. Our model is the parameter $a = 3/5$:

$$F_a(z) = 1 - \frac{1}{10z} - \frac{3}{10z^2}, \qquad 0 \longmapsto \infty \longmapsto 1 \longmapsto a \longmapsto 0, \tag{5.3}$$

with other critical point -6 and other critical value $v_a = v(3/5) = 121/120$. On the sphere marked by $B_a = \{0, \infty, 1, a, v_a\}$ draw the round circles

$$C = \left\{ |z| = \tfrac{241}{240} \right\}, \qquad \eta = \left\{ \left| z - \tfrac{4}{5} \right| = \tfrac{1}{4} \right\}. \tag{5.4}$$

The exterior of C contains exactly the marked points ∞, v_a (note $v_a = 242/240$), and the interior of η contains exactly $a, 1, v_a$ (the distances from $4/5$ to a, 1, v_a are $1/5$, $1/5$, $25/120$, all less than $1/4$, while $|0 - 4/5| > 1/4$). The two circles intersect. Our model curve is

$$\boxed{\gamma_a = T_\eta^2(C),} \tag{5.5}$$

with the sign of Convention 4.1.

Construction 5.1 (Transport to a general map)**.** *Let f be a map of the family with normalized form F_s, $s \in U$, the normalizing Möbius transformation being μ (so $\mu f \mu^{-1} = F_s$). Choose a path s_t, $0 \le t \le 1$, in U from a to s. The five points $0, \infty, 1, s_t, v(s_t)$ stay distinct, so by isotopy extension there is an ambient isotopy H_t of $\mathbb{P}^1$ with $H_0 = \mathrm{id}$ fixing $0, \infty, 1$ and with $H_t(a) = s_t$, $H_t(v_a) = v(s_t)$. Put $\gamma = \mu^{-1}(H_1(\gamma_a)) \subset \mathbb{P}^1 \setminus B$; equivalently, $H_1(\gamma_a)$ is the image of $H_1(C)$ under two twists about $H_1(\eta)$.*

The isotopy class of γ relative to B depends on the path s_t only through its homotopy class in U; Theorem 5.2 holds for every choice.

Theorem 5.2. *Let f, A, B be as in Section 3.2, and let γ be the curve of Construction 5.1. Then the Dehn twist T_γ preserves $\mathcal{D}_f$. If D_0 denotes the connected component of $\mathcal{D}_f$ containing the base point τ_0, then the components $T_\gamma^n(D_0)$, $n \in \mathbb{Z}$, are pairwise distinct.*

Sections 6–9 prove the theorem for the model $f = F_a$ and $\gamma = \gamma_a$; Section 10 deduces the general case.

6 A finite cover of the moduli space

Recall Y and M from (4.1); M is the moduli space of B_a, with $0, \infty, 1$ normalized and (y, z) the positions of the points corresponding to a and v_a. A point of $\mathcal{T}(B_a)$ determines, through (3.1), a quadratic rational map with marked source and target points; the normalized positions of the fourth source and target points give a map from $\mathcal{T}(B_a)$ to a surface $\mathcal{W}$ of marked maps, a finite cover of

M. This is the parameter surface of Hironaka and Koch [HKa, §§4.1–4.2]; in general such spaces are Hurwitz spaces, finite coverings of the moduli space of the target [Ko, Thm. 2.6], [KPS, §3.1].

Consider a quadratic rational map with marked source points $0, \infty, 1, x$ and marked target points $0, \infty, 1, y, z$ realizing the portrait

$$0 \overset{2}{\longmapsto} \infty, \qquad \infty \longmapsto 1, \qquad 1 \longmapsto y, \qquad x \longmapsto 0, \tag{6.1}$$

where the 2 indicates a critical point, and having z as its other critical value. A degree-two map with a double pole at 0 and value 1 at ∞ is $(t-x)(t-r)/t^2$ with zeros x and r, and the condition $1 \mapsto y$ determines r. Hence the map is unique:

$$F_{x,y}(t) = \frac{(t-x)(t-r)}{t^2}, \qquad r = \frac{x+y-1}{x-1}. \tag{6.2}$$

Its critical points are 0 and $2xr/(x+r)$, and the critical value at the latter is $-(x-r)^2/(4xr)$, which in terms of x, y reads

$$Z(x,y) = \frac{((x-1)^2-y)^2}{4x(x+y-1)(1-x)}. \tag{6.3}$$

The marked points must be distinct and the map must have degree two ($r \neq 0$), so the parameter surface and its projection to M are

$$\mathcal{W} = \{(x,y) \in Y^2 : x+y \neq 1,\ Z(x,y) \notin \{0,1,y\}\}, \tag{6.4}$$

$$p\colon \mathcal{W} \longrightarrow M, \qquad p(x,y) = (y, Z(x,y)). \tag{6.5}$$

The maps F_s of the family (3.3) are the marked maps whose fourth source and target points agree: indeed $F_{s,s} = F_s$, the two expressions being quadratic maps with the same poles, zeros and value at ∞, and $Z(s,s) = v(s)$. The conditions $Z(s,s) \neq 0, 1, s$ are precisely the three quadratic factors in (5.2), so the *diagonal* of $\mathcal{W}$ is

$$\mathcal{V} = \mathcal{W} \cap \{x = y\} = \{(s,s) : s \in U\}, \tag{6.6}$$

a connected Riemann surface, since U is connected.

Lemma 6.1. *1. The map p is an unramified covering of degree four, and $\mathcal{W}$ is connected.*

2. For the model (5.3)*, with $\pi_A\colon \mathcal{T}(A_a) \to Y$ and $\pi_B\colon \mathcal{T}(B_a) \to M$ the projections of Section 4, the map*

$$q = (\pi_A \circ F_a^*,\ \pi_A \circ i_*)\colon \mathcal{T}(B_a) \longrightarrow \mathcal{W}$$

is well defined and satisfies $p \circ q = \pi_B$ and $q(\tau_0) = (a,a)$, and it is the universal covering of $\mathcal{W}$. Thus π_B factors through $\mathcal{W}$, and, with $\mathrm{pr}_x, \mathrm{pr}_y$ the two coordinate projections of $\mathcal{W}$, by construction

$$\pi_A \circ F_a^* = \mathrm{pr}_x \circ q, \qquad \pi_A \circ i_* = \mathrm{pr}_y \circ q. \tag{6.7}$$

Proof. (1) Fix $(y,z) \in M$. The equation $Z(x,y) = z$ is

$$((x-1)^2 - y)^2 + 4zx(x-1)(x+y-1) = 0,$$

a monic polynomial equation of degree four in x. Its values

> **Reviewer 2**
>
> The values of what exactly?

at $x = 0, 1, 1-y$ are $(1-y)^2$, y^2 and $y^2(y-1)^2$, all nonzero, so every root satisfies $x \in Y$ and $x + y \neq 1$; since $z \notin \{0, 1, y\}$, every root gives a point of $\mathcal{W}$. Thus $p^{-1}(y, z)$ consists of the roots of this equation. The roots are distinct because of the factorizations

$$\begin{aligned} Z - 1 &= -\frac{(x^2 + y - 1)^2}{4x(x-1)(x+y-1)}, \\ Z - y &= -\frac{(x^2 + 2xy - 2x - y + 1)^2}{4x(x-1)(x+y-1)}, \\ \frac{\partial Z}{\partial x} &= -\frac{(x^2 + y - 1)\left((x-1)^2 - y\right)(x^2 + 2xy - 2x - y + 1)}{4x^2(x-1)^2(x+y-1)^2}, \end{aligned}$$

which show that $\partial Z/\partial x$ vanishes only where $Z = 1$, $Z = 0$ or $Z = y$, and these values of z are excluded in M. So p has exactly four preimages over each point, $\partial Z/\partial x \neq 0$ on $\mathcal{W}$, and the implicit function theorem shows that p is a local homeomorphism. A local homeomorphism between Hausdorff spaces all of whose fibres have the same finite cardinality is a covering (intersect the images of disjoint neighbourhoods of the four preimages of a point, and count), so p is a covering of degree four. Finally $\mathcal{W}$ is the complement in $\mathbb{C}^2$ of the finitely many algebraic curves listed in (6.4), and such a complement is path connected: join two points by two complex lines through a generic auxiliary point, each meeting the curves in finitely many points, which a path in the line can avoid.

(2) Let $\tau = [\phi] \in \mathcal{T}(B_a)$, with ϕ normalized to fix $0, \infty, 1$, and let $F \circ \psi = \phi \circ F_a$ be a pullback diagram (3.1) with ψ also normalized to fix $0, \infty, 1$ (possible since ψ is unique up to a Möbius transformation). Put $x = \psi(a)$, $y = \phi(a)$, $z = \phi(v_a)$. Applying the diagram to the points of A_a and to the critical points $0, -6$ of F_a shows that F has the portrait (6.1) with marked source points $\{0, \infty, 1, x\}$, marked target points $\{0, \infty, 1, y, z\}$, and other critical value $F(\psi(-6)) = z$. Hence $F = F_{x,y}$ and $Z(x, y) = z$ by the uniqueness statement above; $x, y \in Y$ and $z \notin \{0, 1, y\}$ because ψ, ϕ are injective, and $x + y \neq 1$ because F has degree two. Therefore $(x, y) = (\pi_A F_a^* \tau, \pi_A i_* \tau) \in \mathcal{W}$ and $p(x, y) = (y, z) = \pi_B(\tau)$. For τ_0 we may take $\psi = \phi = \mathrm{id}$, so $q(\tau_0) = (a, a)$; and q is continuous because F_a^*, i_*, π_A are.

It remains to see that q is a covering. Let $N \subset M$ be an open ball evenly covered by both p and π_B, with $p^{-1}(N) = \bigsqcup_j N_j$ and $\pi_B^{-1}(N) = \bigsqcup_k \widetilde{N}_k$. The connected set $q(\widetilde{N}_k)$ lies in a single N_j, where $q|_{\widetilde{N}_k} = (p|_{N_j})^{-1} \circ \pi_B|_{\widetilde{N}_k}$ is a homeomorphism onto N_j. Hence each N_j is evenly covered by q (possibly with empty preimage), so $q(\mathcal{T}(B_a))$ is open and closed in the connected space $\mathcal{W}$, and q is a surjective covering; as $\mathcal{T}(B_a)$ is simply connected, it is the universal covering. □

7 When a loop preserves a component

We now set up the covering-space criterion of Hironaka and Koch [HK, §§2.8–2.11] in the coordinates of Lemma 6.1. Let $b = (a, a) = q(\tau_0)$. A loop g in $\mathcal{W}$ based at b determines, via the loop $p \circ g$ in M and (4.2), a deck transformation δ of π_B, which sends τ_0 to the endpoint τ_1 of the lift of g to $\mathcal{T}(B_a)$ starting at τ_0. It is also a deck transformation of q: $q \circ \delta$ and q are both lifts of π_B through p, and they agree at τ_0 because $q(\tau_1) = b$. We denote δ by the same letter g and write $g(\tau)$, as in Section 4. The two *coordinate loops* of g are the loops $\mathrm{pr}_x \circ g$ and $\mathrm{pr}_y \circ g$ in Y based at a.

Lemma 7.1. *Let $E = \operatorname{im}\bigl(\pi_1(\mathcal{V}, b) \to \pi_1(\mathcal{W}, b)\bigr)$ be the set of classes of loops in $\mathcal{W}$ that are homotopic in $\mathcal{W}$ to a loop in the diagonal $\mathcal{V}$. Let $g \in \pi_1(\mathcal{W}, b)$, acting on $\mathcal{T}(B_a)$ as above, and let D_0 be the connected component of $\mathcal{D}_{F_a}$ containing τ_0.*

1. *If the two coordinate loops of g have equal classes in $\pi_1(Y, a)$, then $g(\mathcal{D}_{F_a}) = \mathcal{D}_{F_a}$.*
2. *$g(D_0) = D_0$ if and only if $g \in E$.*

Consequently, if the coordinate loops of g are homotopic and $g^n \notin E$ for every $n \neq 0$, then g preserves $\mathcal{D}_{F_a}$ and the components $g^n(D_0)$, $n \in \mathbb{Z}$, are pairwise distinct.

Proof. (1) By (6.7), F_a^* and i_* are lifts through the universal covering $\pi_A\colon \mathcal{T}(A_a) \to Y$ of the maps $\mathrm{pr}_x \circ q$ and $\mathrm{pr}_y \circ q$, and they take the same value at τ_0 because $\tau_0 \in \mathcal{D}_{F_a}$. Since g is a deck transformation of q, the map $F_a^* \circ g$ is another lift of $\mathrm{pr}_x \circ q$, so $F_a^* \circ g = \delta_x \circ F_a^*$ for a deck transformation δ_x of π_A. To identify δ_x, lift g to a path $\tilde{g}$ from τ_0 to $g(\tau_0)$; then $F_a^* \circ \tilde{g}$ is the lift, starting at $F_a^*(\tau_0)$, of the loop $\mathrm{pr}_x \circ g$, so δ_x is the deck transformation of π_A determined by the class of $\mathrm{pr}_x \circ g$ relative to the base point $F_a^*(\tau_0)$. Likewise $i_* \circ g = \delta_y \circ i_*$ with δ_y determined by the class of $\mathrm{pr}_y \circ g$ relative to the same base point $i_*(\tau_0) = F_a^*(\tau_0)$. Equal classes give $\delta_x = \delta_y =: \delta$, and then for $\tau \in \mathcal{D}_{F_a}$,

$$F_a^*(g\tau) = \delta F_a^*(\tau) = \delta\, i_*(\tau) = i_*(g\tau),$$

so $g(\mathcal{D}_{F_a}) \subset \mathcal{D}_{F_a}$. The coordinate loops of g^{-1} are the inverses of those of g, hence also homotopic, so $g^{-1}(\mathcal{D}_{F_a}) \subset \mathcal{D}_{F_a}$ as well.

(2) First, $\mathcal{D}_{F_a} \subset q^{-1}(\mathcal{V})$: if $F_a^*\tau = i_*\tau$, the two coordinates of $q(\tau)$ agree by (6.7). Next, on a connected component N of $q^{-1}(\mathcal{V})$, the maps $F_a^*|_N$ and $i_*|_N$ are lifts through π_A of one and the same map $\tau \mapsto \mathrm{pr}_x\, q(\tau) = \mathrm{pr}_y\, q(\tau)$, so by uniqueness of lifts they agree on all of N if they agree at one point. Hence $\mathcal{D}_{F_a}$ is a union of connected components of $q^{-1}(\mathcal{V})$; these are path connected, $q\colon q^{-1}(\mathcal{V}) \to \mathcal{V}$ being a covering of the connected manifold $\mathcal{V}$, and D_0 is the component through τ_0.

The deck transformation g permutes the components of $q^{-1}(\mathcal{V})$, so $g(D_0) = D_0$ if and only if there is a path in D_0 from τ_0 to $g(\tau_0)$. Given such a path, its image under q is a loop in $\mathcal{V}$ based at b, homotopic in $\mathcal{W}$ to g because both lift to paths in the simply connected space $\mathcal{T}(B_a)$ with the same endpoints; so $g \in E$. Conversely, if g is homotopic in $\mathcal{W}$ to a loop g' in $\mathcal{V}$, the lift of g' starting at τ_0 stays in $q^{-1}(\mathcal{V})$, hence in D_0, and ends at $g(\tau_0)$ by homotopy lifting; so $g(\tau_0) \in D_0$.

Finally, if $g^j(D_0) = g^k(D_0)$ with $j \neq k$, then $g^{j-k}(D_0) = D_0$, so $g^{j-k} \in E$ by (2), contrary to hypothesis. □

8 Loops in $\mathcal{W}$ representing the twist about γ_a

The goal of this section is a loop g in $\mathcal{W}$ with $P(p \circ g) = T_{\gamma_a}^{\pm 1}$ and trivial coordinate loops (Proposition 8.3). We work in the fibre $\mathcal{W}_a = \{x \in \mathbb{C} : (x, a) \in \mathcal{W}\}$ over $y = a = 3/5$, along which p fixes the four points of A_a and moves only v_a, so that mapping classes are point pushes of v_a. By (6.4) and the factorizations in the proof of Lemma 6.1, the points of $\mathbb{C}$ excluded from $\mathcal{W}_a$ are

$$0,\ 1,\ \tfrac{2}{5}, \qquad 1 \pm \sqrt{\tfrac{3}{5}}\ \ (Z = 0), \qquad \pm\sqrt{\tfrac{2}{5}}\ \ (Z = 1), \qquad \tfrac{2}{5} \pm \tfrac{i\sqrt{6}}{5}\ \ (Z = a); \tag{8.1}$$

here $2/5 = 1 - a$ is the pole of $Z(\cdot, a)$. Their moduli are 0, 1, 0.4, 1.77…, 0.22…, and $\sqrt{2/5} = 0.632\ldots$ for the last four; none is $a = 0.6$, so $\ell(\theta) = (ae^{2\pi i\theta}, a)$, $0 \le \theta \le 1$, is a loop in $\mathcal{W}$ based at b. Its image $p \circ \ell$ in M moves v_a along the loop $\zeta(\theta) = Z(ae^{2\pi i\theta}, a)$ in $\mathbb{P}^1 \setminus A_a$ based at $v_a = Z(a, a)$.

Let m be a positive meridian about the pole $x = 2/5$ in $\mathcal{W}_a$: travel from a along the real interval toward $2/5$, go once counterclockwise around a small circle centred at $2/5$, and return along the interval.

Lemma 8.1. *The mapping class $P(p \circ m)$ is $T_C^{\pm 1}$, a single Dehn twist about the round circle C of* (5.4).

Proof. On $(2/5, 3/5]$ the factorizations give $\partial Z(x,a)/\partial x < 0$, because in the numerator of $\partial Z/\partial x$ the factors $x^2 - 2/5$ and $(x-1)^2 - 3/5$ are negative and $x^2 - \frac{4}{5}x + \frac{2}{5}$ is positive. Moreover $Z(a,a) = v_a$, so $Z(x,a) \geq v_a > 1$ on $(2/5, a]$ by monotonicity; the pole of $Z(\cdot, a)$ at $2/5$ is simple, and $Z(x,a) \to +\infty$ as $x \downarrow 2/5$. Thus $p \circ m$ moves v_a out along the positive real ray, once around ∞ (a small circle about a simple pole maps to a loop winding once about ∞), and back. For a small enough circle this motion takes place in the closed disk $\Delta = \{|z| \geq 241/240\} \cup \{\infty\}$ bounded by C, whose only marked points are v_a and ∞; in $\Delta \setminus \{\infty\}$, whose fundamental group is $\mathbb{Z}$, the loop is homotopic to a simple loop once around ∞, so (4.3) gives $P(p \circ m) = T_C^{\pm 1}$. □

The homotopy class of ζ, not merely its homology class, must be identified, since the isotopy class of γ_a depends on it.

Lemma 8.2. *The loop ζ is homotopic, as a loop based at v_a in $\mathbb{P}^1 \setminus A_a$, to two clockwise circuits around the segment $[a, 1]$ inside the disk bounded by η. Consequently $h := P(\zeta^{-1})$ satisfies $h(C) = T_\eta^2(C) = \gamma_a$, with the sign of Convention 4.1.*

Proof. Step 1: enlarging the circle. Put $R = \sqrt{1-a} = \sqrt{2/5} > a$. No excluded point in (8.1) has modulus in $[a, R)$, so for $a \leq r < R$ the loops $\zeta_r(\theta) = Z(re^{2\pi i\theta}, a)$ avoid $0, a, 1, \infty$ and depend continuously on r. Their base points $Z(r,a)$ move along the real path $\sigma_r = Z([a,r], a) \subset (1, v_a]$, since $Z(\cdot, a)$ is decreasing on $(2/5, R)$ (the sign analysis in the proof of Lemma 8.1 is valid there) with $Z(R,a) = 1$. Hence ζ is based-homotopic in $\mathbb{P}^1 \setminus A_a$ to $\sigma_r^{-1}\zeta_r\sigma_r$ for every $r \in [a, R)$.

Step 2: the circle $|x| = R$ maps onto $[a, 1]$, and nothing inside it does. The substitution $u = x + (1-a)/x$ (a Joukowski map) simplifies Z: writing $(x-1)^2 - a = x(u-2)$ and $(x+a-1)(1-x) = x(2-a-u)$ gives

$$Z(x,a) = G(u), \qquad G(u) = \frac{(u-2)^2}{4(2-a-u)}. \tag{8.2}$$

For $x = Re^{i\vartheta}$ we have $u = 2R\cos\vartheta$, so on $|x| = R$ the variable u runs over $[-2R, 2R]$, where $2 - a - u \geq 2 - a - 2R > 0$ and

$$G(u) - 1 = \frac{u^2 - 4(1-a)}{4(2-a-u)}, \qquad G(u) - a = \frac{(u - 2(1-a))^2}{4(2-a-u)}.$$

Thus on $[-2R, 2R]$ the function G takes values in $[a, 1]$, equals 1 exactly at the endpoints, and equals a exactly at $u = 2(1-a)$; since $G'(u) = (u-2)(2-2a-u)/(4(2-a-u)^2)$, G decreases from 1 to a and then increases back to 1. Hence for $z \in (a, 1)$ the equation $G(u) = z$ has two distinct solutions $u \in (-2R, 2R)$, and for each of them the two roots of $x^2 - ux + (1-a) = 0$ are non-real conjugates with $|x|^2 = 1 - a$, hence lie on $|x| = R$. This produces four distinct points of $|x| = R$ mapping to z, which exhaust the four preimages of (a, z) under p. For the endpoints the factorizations of $Z - 1$ and $Z - a$ show that all preimages of 1 and of a (namely $\pm R$ and $2/5 \pm i\sqrt{6}/5$, with multiplicity) also lie on $|x| = R$. So $Z(\{|x| = R\}, a) = [a, 1]$, and no x with $|x| < R$ maps into $[a, 1]$.

Step 3: the homotopy class. The segment $[a,1]$ lies in the open disk bounded by η, so by Step 2 and continuity, for $r<R$ close to R the loop ζ_r lies in that disk and avoids $[a,1]$; the base point path $\sigma_r\subset(1,v_a]$ also lies in the disk and avoids $[a,1]$. So $\sigma_r^{-1}\zeta_r\sigma_r$ is a loop in the η-disk minus the segment $[a,1]$. This space is an annulus, with fundamental group $\mathbb{Z}$ generated by a counterclockwise circuit around $[a,1]$, and the class of a loop is its winding number about any point of the segment, say about 1. Since the homotopy of Step 1 avoids 1, this winding number equals $\mathrm{wind}_1(\zeta)$, which we compute by the argument principle applied to

$$Z(x,a)-1=-\frac{(x^2+a-1)^2}{4x(x-1)(x+a-1)}$$

on the disk $|x|<a$: the numerator vanishes only at $\pm R$, outside the disk, while the denominator has the two simple zeros 0 and $1-a=2/5$ inside it. Hence $\mathrm{wind}_1(\zeta)=0-2=-2$, and ζ is based-homotopic to two clockwise circuits around $[a,1]$ in the η-disk, as claimed.

Step 4: the point push. Choose a round circle κ enclosing $a,1$ but not v_a and lying inside both C and η; for instance $\kappa=\{|z-\frac{4}{5}|=\frac{97}{480}\}$, for which $|a-\frac{4}{5}|=|1-\frac{4}{5}|=\frac{96}{480}<\frac{97}{480}<\frac{100}{480}=|v_a-\frac{4}{5}|$, and the largest modulus on κ is $\frac{481}{480}<\frac{241}{240}$. The annulus between κ and η contains the single marked point v_a, and by Step 3, ζ^{-1} is homotopic to two counterclockwise circuits of v_a around this annulus. By Convention 4.1 one such circuit has point push $T_\eta T_\kappa^{-1}$, so, the twists about the disjoint curves κ and η commuting,

$$h=P(\zeta^{-1})=T_\eta^2T_\kappa^{-2}.$$

Since κ is disjoint from C, the twist T_κ fixes the isotopy class of C, and $h(C)=T_\eta^2(C)=\gamma_a$. □

Proposition 8.3. *Let*

$$g=\ell^{-1}m\ell\in\pi_1(\mathcal{W},b),\tag{8.3}$$

the loop that traverses ℓ, then m, then ℓ backwards. Then $P(p\circ g)=T_{\gamma_a}^{\pm1}$: the deck transformation g acts on $\mathcal{T}(B_a)$ as the Dehn twist about γ_a or its inverse. Both coordinate loops of g are null-homotopic in Y. In particular T_{γ_a} preserves $\mathcal{D}_{F_a}$.

Proof. Since $p\circ\ell$ moves only v_a, along ζ, we have $P(p\circ\ell)=P(\zeta)=h^{-1}$ with h as in Lemma 8.2. Multiplicativity of P (Section 4) and Lemma 8.1 give

$$P(p\circ g)=P(p\circ\ell^{-1})\,P(p\circ m)\,P(p\circ\ell)=h\,T_C^{\pm1}\,h^{-1}=T_{h(C)}^{\pm1}=T_{\gamma_a}^{\pm1},$$

using (P2) and Lemma 8.2. The y-coordinate of g is constant. The x-coordinate of m is a small loop around $2/5$ together with a real approach path in $(2/5,3/5]$; this loop lies in the disk $|x-\frac{1}{2}|<\frac{1}{2}$, which avoids 0 and 1, so it is null-homotopic in Y; hence the x-coordinate loop of g is homotopic to $(\mathrm{pr}_x\circ\ell)^{-1}(\mathrm{pr}_x\circ\ell)$, which is trivial. Lemma 7.1(1) now shows that g, hence T_{γ_a}, preserves $\mathcal{D}_{F_a}$. □

9 No nonzero power of the twist preserves the base component

By Lemma 7.1 and Proposition 8.3, it remains to show that $g^n\notin E$ for $n\neq0$. Consider

$$\Phi\colon\mathcal{W}\longrightarrow Y\times\mathbb{C}^*,\qquad\Phi(x,y)=\big(t(x,y),L(x,y)\big)=\left(\frac{xy}{(1-x)(1-y)},\ (1-x)(1-y)\right).\tag{9.1}$$

It is well defined: $L \neq 0$ and $t \neq 0$ since $x, y \in Y$, and $t - 1 = (x + y - 1)/((1 - x)(1 - y)) \neq 0$. In terms of the symmetric functions $e_1 = x + y$, $e_2 = xy$ one has $L = 1 - e_1 + e_2$ and $t = e_2/L$, so Φ is the symmetric quotient $(x, y) \mapsto (x + y, xy)$ of Hironaka–Koch [HK, §2.12], restricted to $\mathcal{W}$, up to this change of coordinates.

The constraint on diagonal loops. Enlarge the diagonal by filling in its six quadratic punctures: $\widehat{\mathcal{V}} = \{(s, s) : s \in \mathbb{C} \setminus \{0, 1, 1/2\}\} \supset \mathcal{V}$, on which (9.1) still makes sense, so that the covering (9.3) below covers all of Y. In the coordinate $w = s/(1 - s)$, which maps $\mathbb{C} \setminus \{0, 1, 1/2\}$ bijectively onto $\mathbb{C} \setminus \{0, 1, -1\}$, the restriction of Φ is

$$t = w^2, \qquad L = (1 + w)^{-2}, \qquad w \in \mathbb{C} \setminus \{0, 1, -1\}, \tag{9.2}$$

because $1 + w = 1/(1 - s)$. The squaring map

$$\mathbb{C} \setminus \{0, 1, -1\} \longrightarrow Y, \qquad w \longmapsto w^2, \tag{9.3}$$

is a two-sheeted covering; our base points are $w_0 = a/(1 - a) = 3/2$ and $t_0 = w_0^2 = 9/4$.

> **Reviewer 2**
> The previous coordinate change and the following lemma seem like an unnecessary detour. Why not do the same computations in the original coordinates?

Lemma 9.1. *Let λ be a loop in $\mathcal{V}$ based at b. Then $w \circ \lambda$ is the lift of the loop $t \circ \lambda$ through* (9.3) *starting at w_0; it is closed, and*

$$\operatorname{wind}_0(L \circ \lambda) = -2 \operatorname{wind}_{-1}(w \circ \lambda). \tag{9.4}$$

The based homotopy class of $w \circ \lambda$ in $\mathbb{C} \setminus \{0, 1, -1\}$, hence the right-hand side of (9.4)*, depends only on the based homotopy class of $t \circ \lambda$ in Y.*

Proof. On $\mathcal{V} \subset \widehat{\mathcal{V}}$ formula (9.2) applies: $w \circ \lambda$ is a closed path in $\mathbb{C} \setminus \{0, 1, -1\}$ from w_0 with $(w \circ \lambda)^2 = t \circ \lambda$, hence the lift of $t \circ \lambda$ from w_0, and $L \circ \lambda = (1 + w \circ \lambda)^{-2}$ gives (9.4). The last assertion is homotopy lifting for the covering (9.3). □

The loops g^n violate the constraint. On the fibre $y = a$ the map (9.1) becomes

$$t = \frac{3x}{2(1 - x)}, \qquad L = \frac{2}{5}(1 - x). \tag{9.5}$$

Let $\alpha = t \circ \ell$ and $\beta = t \circ m$, loops in Y based at t_0.

Lemma 9.2. *For every integer n, the loop g^n satisfies*

$$\operatorname{wind}_0(L \circ g^n) = 0, \qquad \operatorname{wind}_{-1}\big(\widetilde{t \circ g^n}\big) = n, \tag{9.6}$$

where $\widetilde{t \circ g^n}$ is the lift of $t \circ g^n$ through (9.3) *starting at w_0, which is a closed loop.*

Proof. The loop ℓ. Along ℓ we have $x = ae^{2\pi i\theta}$ with $|x| = a < 1$, so $1 - x$ does not wind about 0 and (9.5) gives $\operatorname{wind}_0(\alpha) = 1$ and $\operatorname{wind}_0(L \circ \ell) = 0$. Since $\operatorname{wind}_0(\alpha)$ is odd, continuation of the square root along α, or along α^{-1}, interchanges the two sheets of (9.3): the lift $\tilde{\alpha}$ of α starting at w_0 ends at $-w_0$.

The loop m. The Möbius map $x \mapsto t$ in (9.5) sends $2/5$ to 1 and a to $9/4$, is increasing on the real interval $(2/5, a]$, which it maps onto $(1, 9/4]$, and, being conformal, sends a small counterclockwise circle about $2/5$ to a small counterclockwise loop about 1. Hence β is a positive meridian about 1 in Y, approached from t_0 along the real axis to the right of 1. Its lift starting at $w_0 = 3/2$ runs along $(1, 3/2]$ and around a small loop about $+1$, so it is a positive meridian about $+1$; the lift starting at $-w_0$ is its negative, a positive meridian about -1 (the map $w \mapsto -w$ preserves orientation). Also $\mathrm{wind}_0(L \circ m) = 0$, since $1 - x$ stays near the positive interval $[2/5, 3/5]$ along m.

The loop g. By additivity, $\mathrm{wind}_0(L \circ g) = -\mathrm{wind}_0(L \circ \ell) + \mathrm{wind}_0(L \circ m) + \mathrm{wind}_0(L \circ \ell) = 0$, and so $\mathrm{wind}_0(L \circ g^n) = 0$. The t-projection of g is $\alpha^{-1}\beta\alpha$: first α, then β, then α backwards. Its lift from w_0 consists of $\tilde{\alpha}$, ending on the other sheet at $-w_0$, then the lift of β from $-w_0$, a positive meridian about -1, then $\tilde{\alpha}$ reversed, returning to w_0. The contributions of $\tilde{\alpha}$ and its reverse to the change of argument of $w + 1$ cancel, so this closed lift has winding number 1 about -1, and the lift of $t \circ g^n = (\alpha^{-1}\beta\alpha)^n$ from w_0, the n-fold concatenation, has winding number n. □

Proposition 9.3. *For every $n \neq 0$, $g^n \notin E$. Consequently Theorem 5.2 holds for the model map F_a and the curve $\gamma_a = T_\eta^2(C)$.*

Proof. Suppose $g^n \in E$, so that g^n is homotopic in $\mathcal{W}$ to a loop λ in $\mathcal{V}$ based at b. Applying Φ, the loops $t \circ \lambda$ and $t \circ g^n$ are based-homotopic in Y, and $L \circ \lambda$ and $L \circ g^n$ are homotopic in $\mathbb{C}^*$. By Lemma 9.1, the lift $w \circ \lambda$ is based-homotopic to the lift of $t \circ g^n$ used in Lemma 9.2, so $\mathrm{wind}_{-1}(w \circ \lambda) = n$, while $\mathrm{wind}_0(L \circ \lambda) = \mathrm{wind}_0(L \circ g^n) = 0$. The constraint (9.4) then reads $0 = -2n$, so $n = 0$.

For the second assertion, g acts as $T_{\gamma_a}^{\pm 1}$ and has homotopic coordinate loops (Proposition 8.3), so by Lemma 7.1 the twist T_{γ_a} preserves $\mathcal{D}_{F_a}$ and the components $T_{\gamma_a}^n(D_0) = g^{\pm n}(D_0)$, $n \in \mathbb{Z}$, are pairwise distinct. □

Remark 9.4. The same computation applies to the other conjugate $g' = \ell m \ell^{-1}$ (its t-projection $\alpha\beta\alpha^{-1}$ also changes sheet before the meridian, as α^{-1} has odd winding number about 0, and again $\mathrm{wind}_0(L \circ g') = 0$), whose mapping class is $h^{-1}T_C^{\pm 1}h = T_{h^{-1}(C)}^{\pm 1}$ with $h^{-1}(C) = T_\eta^{-2}T_\kappa^2(C) = T_\eta^{-2}(C)$. Hence Theorem 5.2 holds for both curves $T_\eta^{\pm 2}(C)$.

10 Transport to an arbitrary map of the family

> **Reviewer 2**
>
> All statements in this section are quite straightforward; this can be shortened substantially.

Let f, μ, F_s, s_t, H_t be as in Construction 5.1. We show that changing markings by $H = H_1$ carries $\mathcal{D}_{F_a}$ onto $\mathcal{D}_{F_s}$, D_0 to the base component of $\mathcal{D}_{F_s}$, and conjugates T_{γ_a} to $T_{H(\gamma_a)}$.

Lemma 10.1. *There is a continuous family of homeomorphisms K_t of $\mathbb{P}^1$, $0 \le t \le 1$, with $K_0 = \mathrm{id}$ and*

$$F_{s_t} \circ K_t = H_t \circ F_a \qquad (0 \le t \le 1). \tag{10.1}$$

It satisfies $K_t|_{A_a} = H_t|_{A_a}$ for all t; consequently $H_t^{-1}K_t$ is an isotopy from the identity through homeomorphisms fixing A_a pointwise, and K_1 is isotopic to H_1 relative to A_a.

Proof. Every F_s is Möbius-equivalent to $Q(z) = z^2$, with Möbius maps depending continuously on s: completing the square in $F_s(z) = 1 + b_s/z + c_s/z^2$ and recalling $v(s) = 1 - b_s^2/(4c_s)$ gives $F_s = v(s) + c_s\, J_s^2$, that is,

$$\nu_s \circ F_s = Q \circ J_s, \qquad J_s(z) = \frac{1}{z} + \frac{b_s}{2c_s}, \qquad \nu_s(w) = \frac{w - v(s)}{c_s},$$

with $c_s \neq 0$ for $s \in U$. The isotopy $\widehat{H}_t = \nu_{s_t} \circ H_t \circ \nu_a^{-1}$ starts at the identity and fixes 0 and ∞, because $H_t(v_a) = v(s_t)$ and $H_t(\infty) = \infty$. By the homotopy lifting property of the covering $Q\colon \mathbb{C}^* \to \mathbb{C}^*$, the homotopy $(z,t) \mapsto \widehat{H}_t(Q(z))$ lifts uniquely to a continuous $\rho\colon \mathbb{C}^* \times [0,1] \to \mathbb{C}^*$ with $\rho_0 = \mathrm{id}$ and $Q \circ \rho_t = \widehat{H}_t \circ Q$; lifting $\widehat{H}_t^{-1} \circ Q$ in the same way gives ρ_t' with $\rho_t'\rho_t = \mathrm{id} = \rho_t\rho_t'$ (both products lift the constant homotopy Q and start at id), so each ρ_t is a homeomorphism of $\mathbb{C}^*$, and ρ extends continuously to $\mathbb{P}^1 \times [0,1]$ by $\rho_t(0) = 0$, $\rho_t(\infty) = \infty$, since $|\rho_t(z)|^2 = |\widehat{H}_t(z^2)|$ and $\widehat{H}$ is continuous on the compact space $\mathbb{P}^1 \times [0,1]$. Then $K_t = J_{s_t}^{-1} \circ \rho_t \circ J_a$ is a continuous family of homeomorphisms with $K_0 = \mathrm{id}$ and $F_{s_t}K_t = \nu_{s_t}^{-1} Q \rho_t J_a = \nu_{s_t}^{-1}\widehat{H}_t Q J_a = H_t \nu_a^{-1} Q J_a = H_t F_a$, which is (10.1).

For the marked points, (10.1) gives $F_{s_t}(K_t(w)) = H_t(F_a(w))$ for $w \in A_a$. Thus $K_t(0)$ is the critical point of F_{s_t} over ∞, namely 0; $K_t(\infty)$ lies in $F_{s_t}^{-1}(1) = \{\infty, -c_{s_t}/b_{s_t}\}$, $K_t(1)$ in $F_{s_t}^{-1}(s_t)$, and $K_t(a)$ in $F_{s_t}^{-1}(0) \ni s_t$. For $s_t \in U$ each of the latter three fibres (over the non-critical values 1, s_t, 0) consists of two distinct points depending continuously on t: $b_{s_t} \neq 0$ since $s_t^2 + s_t - 1 \neq 0$, and s_t, 0 are not critical values of F_{s_t} since $v(s_t) \neq s_t, 0$. By continuity from $t = 0$, $K_t(0) = 0$, $K_t(\infty) = \infty$, $K_t(1) = 1$ and $K_t(a) = s_t$, which are the values of H_t on A_a. The remaining assertions follow. □

Proof of Theorem 5.2. Write $H = H_1$, $K = K_1$. Since H maps B_a onto B_s (notation of Section 5), changing markings by H is a biholomorphic isomorphism

$$\Psi\colon \mathcal{T}(B_a) \longrightarrow \mathcal{T}(B_s), \qquad [\phi] \longmapsto [\phi \circ H^{-1}],$$

and likewise $[\psi] \mapsto [\psi \circ H^{-1}]$ identifies $\mathcal{T}(A_a)$ with $\mathcal{T}(A_s)$, $A_s = \{0, \infty, 1, s\} = H(A_a) = K(A_a)$.

Ψ carries $\mathcal{D}_{F_a}$ onto $\mathcal{D}_{F_s}$. Let ϕ be a marking with pullback diagram $\widetilde{F} \circ \psi = \phi \circ F_a$ for F_a. Using $F_a \circ K^{-1} = H^{-1} \circ F_s$ from (10.1),

$$\widetilde{F} \circ (\psi K^{-1}) = \phi \circ F_a \circ K^{-1} = (\phi H^{-1}) \circ F_s,$$

so ψK^{-1} is a source marking for the target marking ϕH^{-1} in a pullback diagram for F_s: $F_s^*\Psi[\phi] = [\psi K^{-1}]$, while $i_*\Psi[\phi] = [\phi H^{-1}]$. Now $[\psi K^{-1}] = [\phi H^{-1}]$ in $\mathcal{T}(A_s)$ if and only if $[\psi K^{-1} H] = [\phi]$ in $\mathcal{T}(A_a)$, and $K^{-1}H$ is isotopic to the identity relative to A_a by Lemma 10.1, so this holds if and only if $[\psi] = [\phi]$ in $\mathcal{T}(A_a)$, that is, $[\phi] \in \mathcal{D}_{F_a}$. Hence $\Psi(\mathcal{D}_{F_a}) = \mathcal{D}_{F_s}$.

Ψ conjugates twists. For a mapping class $k \in \mathrm{PMod}(B_a)$, $\Psi([\phi] \cdot k) = [\phi k H^{-1}] = [\phi H^{-1}] \cdot (HkH^{-1})$, so Ψ intertwines the action of T_{γ_a} with that of $HT_{\gamma_a}H^{-1} = T_{H(\gamma_a)}$, by the naturality of twists (P2), which holds verbatim for the homeomorphism $H\colon (\mathbb{P}^1, B_a) \to (\mathbb{P}^1, B_s)$.

Ψ carries D_0 to the base component. The path of markings $[H_t]$ in $\mathcal{T}(B_a)$ lies in $\mathcal{D}_{F_a}$: by (10.1), K_t is a source marking for the target marking H_t, and $[K_t] = [H_t]$ in $\mathcal{T}(A_a)$ because $H_t^{-1}K_t$ is isotopic to the identity relative to A_a (Lemma 10.1). The path starts at τ_0, so its endpoint $[H]$ lies in D_0, and $\Psi[H] = [\mathrm{id}]$ is the base point of $\mathcal{T}(B_s)$. Hence $\Psi(D_0)$ is the base component of $\mathcal{D}_{F_s}$, and the components $T_{H(\gamma_a)}^n(\Psi(D_0)) = \Psi(T_{\gamma_a}^n(D_0))$, $n \in \mathbb{Z}$, are pairwise distinct by Proposition 9.3. Finally, conjugating by μ^{-1} identifies $(\mathbb{P}^1, B_s, F_s)$ with $(\mathbb{P}^1, B, f)$ and $H(\gamma_a)$ with γ, which proves the theorem for f. □

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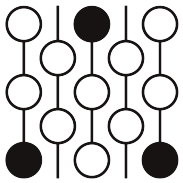
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Review of Problem 7 – ProofCouncil

First Proof Third Batch
Reviewer 3
October 8–9, 2026



1 Review

I am confident that the submission provides a correct solution to the proposed problem. The lack of diagrams or drawings, the poor choice of notation, and the convoluted explanations make it hard to follow the arguments at times; the solution invokes numerous non-obvious coordinate changes that require delicate computations and understanding the geometry behind them is non-trivial.

Up to and including section 7, most of what is done is reproducing the strategy, calculations, and conclusions of the cited papers, and the "novelty work" of the solution is in sections 8, 9 and 10. In some ways, the solution proposed is more direct than the abstract constructions in the cited papers, with clever coordinate changes to avoid discussing generators of various fundamental groups related to the symmetric quotient construction of (Hironaka, Koch). The tradeoff is the sheer amount of computation needed to verify all of the stated facts. There is also a small problem with a double citation, to both the arxiv submission and the journal publication of (Hironaka, Koch).

Correctness.

- 1: essentially flawless mathematical argument
- 0.5: minor errors that are easily corrected
- 0: blank solution, or acknowledged partial result
- -0.5: major errors
- -1: proposed proof makes no progress towards the result.

Note that points should be docked for wrong statements/ solutions.
Score: 1

Exposition.

- 1: well-written and clear solution.
- 0: argument that is possible to follow with serious effort
- -1: very poor exposition

Score: 0

Attributions.

- 1: solutions contains fully appropriate attributions to related work and theorems being used.
- 0: moderate issues with missing attributions or hallucinated references
- -1: blatant plagiarism or hallucinations

Score: 1

2 Submission with inline comments

Abstract

Let f be a quadratic rational map one of whose critical points lies in a cycle A of length four and whose other critical value v lies outside A, and let $B = A \cup \{v\}$. Epstein's deformation space $\mathcal{D}_f \subset \mathcal{T}(\mathbb{S}^2, B)$ is known to be disconnected (Hironaka–Koch). We write down an explicit simple closed curve γ on $(\mathbb{S}^2, B)$, for a normalized model map the image of a round circle under two Dehn twists about a second round circle, transported to a general f along a path in parameter space, and prove that the Dehn twist T_γ itself preserves $\mathcal{D}_f$ and that the translates $T_\gamma^n(D_0)$, $n \in \mathbb{Z}$, of the base component D_0 are pairwise distinct components, using the covering-space framework of Hironaka–Koch and a winding-number obstruction.

3 Introduction

3.1 Deformation spaces

Throughout, $\mathbb{P}^1$ is the Riemann sphere, which we also regard as the oriented topological sphere $\mathbb{S}^2$. For a finite set $B \subset \mathbb{P}^1$ with $|B| \geq 3$, the *Teichmüller space* $\mathcal{T}(B) = \mathcal{T}(\mathbb{S}^2, B)$ is the set of orientation-preserving homeomorphisms $\phi\colon \mathbb{P}^1 \to \mathbb{P}^1$ (*markings*), where $\phi \sim \phi'$ when $\phi' = \mu \circ \phi \circ h$ for a Möbius transformation μ and a homeomorphism h isotopic to the identity relative to B. Pulling back the standard complex structure by ϕ identifies $\mathcal{T}(B)$ with the complex structures on $(\mathbb{S}^2, B)$ up to isotopy relative to B; the class $\tau_0 = [\mathrm{id}]$ is the *base point*.

Let $f\colon \mathbb{P}^1 \to \mathbb{P}^1$ be an orientation-preserving branched cover of degree $d \geq 2$ with set of critical values $V(f)$, and let $A, B \subset \mathbb{P}^1$ be finite sets with $|A|, |B| \geq 3$ and $f(A) \cup V(f) \subset B$. Pulling back the complex structure of a marking ϕ under f gives a complex structure on $(\mathbb{S}^2, A)$; concretely, there are a homeomorphism ψ, unique up to post-composition with a Möbius transformation, and a rational map F with

$$F \circ \psi = \phi \circ f, \tag{3.1}$$

and $f^*[\phi] := [\psi] \in \mathcal{T}(A)$. This is the Thurston pullback map $f^*\colon \mathcal{T}(B) \to \mathcal{T}(A)$; it is holomorphic [BCT, §1.3, Prop. 1.4]. We call ϕ the *target marking* and ψ the *source marking* of the diagram (3.1). If $A \subset B$, forgetting the points of $B \setminus A$ gives the holomorphic *forgetful map* $i_*\colon \mathcal{T}(B) \to \mathcal{T}(A)$, $[\phi] \mapsto [\phi]$. When $f(A) \cup V(f) \cup A \subset B$, both maps are defined, and the *deformation space* of f is their equalizer

$$\mathcal{D}_f = \{\tau \in \mathcal{T}(B) : f^*(\tau) = i_*(\tau)\}.$$

This notion is due to Epstein [Ep]; see [BCT, §1.4] for an exposition. When f is rational, as in the problem considered here, we may take $\psi = \mathrm{id}$, $F = f$ in (3.1) for $\phi = \mathrm{id}$, so $\tau_0 \in \mathcal{D}_f$. (By Epstein's theorem [BCT, Thm. 1.5], under the standing hypotheses above, if f is not a topological $(2, 2, 2, 2)$-map in the sense of [BCT, §1.4], then $\mathcal{D}_f$, whenever nonempty, is a smooth complex submanifold of $\mathcal{T}(B)$ of dimension $|B \setminus A|$. For the rational family considered here $\tau_0 \in \mathcal{D}_f$, and the periodic critical point excludes the exceptional type.)

The *pure mapping class group* $\mathrm{PMod}(B)$, the group of isotopy classes relative to B of orientation-preserving homeomorphisms of $\mathbb{P}^1$ fixing B pointwise, acts on $\mathcal{T}(B)$ on the right by

$$[\phi] \cdot h = [\phi \circ h]; \tag{3.2}$$

for a subset $D \subset \mathcal{T}(B)$ we write $h(D) = \{\tau \cdot h : \tau \in D\}$. A mapping class that preserves $\mathcal{D}_f$ permutes its connected components. For a simple closed curve γ in $\mathbb{P}^1 \setminus B$ we write $T_\gamma \in \mathrm{PMod}(B)$

for the Dehn twist about γ. The question addressed here asks for an explicit γ such that some power of T_γ preserves $\mathcal{D}_f$ and has an infinite orbit on the components.

3.2 The family and the result

Let f be a quadratic rational map with critical points c_1, c_2 such that c_1 lies in a cycle of length 4 and the critical value $v = f(c_2)$ is not in this cycle, and put

$$A = \{c_1, f(c_1), f^{\circ 2}(c_1), f^{\circ 3}(c_1)\}, \qquad B = A \cup \{v\}.$$

Then $f(A) = A$ and $V(f) = \{f(c_1), v\} \subset B$, so $\mathcal{D}_f \subset \mathcal{T}(B)$ is defined; $\mathcal{T}(B)$ has dimension 2. Conjugating f by the Möbius transformation that sends $c_1, f(c_1), f^{\circ 2}(c_1)$ to $0, \infty, 1$ normalizes the cycle to $0 \mapsto \infty \mapsto 1 \mapsto s \mapsto 0$, and Section 5 shows that this forces f to become

$$F_s(z) = 1 + \frac{-s^2 - s + 1}{(s-1)z} + \frac{s(2s-1)}{(s-1)z^2}, \qquad s \in U, \tag{3.3}$$

for an explicit Zariski-open set $U \subset \mathbb{C}$ given in (5.2). Our model is F_a with $a = 3/5$, whose marked set is $B_a = \{0, \infty, 1, a, v_a\}$, where $v_a = 121/120$ is its second critical value (computed in (5.1)).

> **Reviewer 3**
>
> This normal form is already present in the literature (Firsova, Kahn, Selinger); this not necessarily to the detriment of the exposition, making it more self-contained, but a reference would be convenient here.

The answer (Theorem 5.2) is as follows. Let C be the round circle $|z| = 241/240$, which separates $\{\infty, v_a\}$ from $\{0, 1, a\}$, and let η be the round circle $|z - 4/5| = 1/4$, which encloses exactly $a, 1, v_a$. Put $\gamma_a = T_\eta^2(C)$, the image of C under two Dehn twists about η (with the sign of Convention 4.1; either sign works, Remark 9.4). For a general f of the family, let γ be obtained from γ_a by following any path in U from a to the parameter of f and dragging the marked points along. Then T_γ preserves $\mathcal{D}_f$, and the translates $T_\gamma^n(D_0)$, $n \in \mathbb{Z}$, of the component $D_0 \ni \tau_0$ are pairwise distinct components; so the power of the twist can be taken to be 1. (A path is needed: γ_a and C separate B_a in the same way, so separation data do not determine the isotopy class of γ.)

3.3 Relation to earlier work

Hironaka and Koch [HK] proved that for this family $\mathcal{D}_f$ has infinitely many connected components ([HK, Thm. 1.4]), introducing the tools we use: the surface of marked quadratic maps with its degree-four covering of moduli space and its diagonal ([HKa, §§4.1–4.2], [HK, §3]), the description of $\mathcal{D}_f$ through mapping classes that lift under f and through equalizers ([HK, §§2.8–2.11]), and a symmetric quotient of the parameter surface used as an obstruction ([HK, §2.12]); they also exhibit a mapping class preserving $\mathcal{D}_f$ with an infinite orbit on components ([HK, Prop. 3.2]). Here [HKa] is the first arXiv version of their paper and [HK] the revised author version, whose numbering we follow. Firsova, Kahn and Selinger [FKS] treat the same family ([FKS, §6]; numbering of arXiv v4), study twists along multicurves adapted to the pullback ("equalizing multicurves"), and prove that these deformation spaces are not contractible ([FKS, Lemma 8.4]). The present note adds the identification of a single, explicitly drawn Dehn twist with an infinite component orbit, which we have not found in [HK] or [FKS], with an elementary verification; our obstruction map Φ (Section 9) is a form of the symmetric quotient of [HK].

3.4 Strategy and outline

The proof has five steps. (1) Section 6 factors the universal cover $\mathcal{T}(B_a) \to M$ of the moduli space M of five marked points through a finite cover $\mathcal{W} \to M$, the surface of quadratic maps with the prescribed portrait of marked orbits; the pullback and forgetful maps become the coordinate projections of $\mathcal{W}$, and $\mathcal{D}_{F_a}$ lies over the diagonal $\mathcal{V} \subset \mathcal{W}$. (2) Section 7: a loop g in $\mathcal{W}$ acts on $\mathcal{T}(B_a)$ as a mapping class, which preserves $\mathcal{D}_{F_a}$ if the two coordinate loops of g are homotopic, and preserves the base component D_0 if and only if g is homotopic in $\mathcal{W}$ to a loop in $\mathcal{V}$. (3) Section 8 exhibits a loop $g = \ell^{-1} m \ell$ in $\mathcal{W}$ acting as $T_{\gamma_a}^{\pm 1}$: m is a meridian acting as the twist about C, and the point push along ℓ is identified exactly by a Joukowski substitution. (4) Section 9 shows that no power g^n, $n \neq 0$, is homotopic to a loop in $\mathcal{V}$, using a map $\Phi = (t, L) \colon \mathcal{W} \to (\mathbb{C} \setminus \{0,1\}) \times \mathbb{C}^*$ whose first coordinate becomes, on an enlarged diagonal, the two-sheeted covering $w \mapsto w^2$, while its second coordinate imposes a linear relation between two winding numbers that the loops g^n violate. (5) Section 10 transports the conclusion to every map of the family by lifting an isotopy of marked points through the branched covers.

4 Conventions on loops, point pushing and twists

For B as above, the *moduli space* $\mathcal{M}(B)$ is the set of injective maps $B \to \mathbb{P}^1$ modulo post-composition with Möbius transformations. Choosing three points of B and normalizing their images to $0, \infty, 1$ identifies $\mathcal{M}(B)$ with an open subset of $\mathbb{C}^{|B|-3}$. The projection $\pi_B \colon \mathcal{T}(B) \to \mathcal{M}(B)$, $[\phi] \mapsto [\phi|_B]$, is a holomorphic universal covering whose deck group is $\mathrm{PMod}(B)$ acting by (3.2); see [KPS, §2.1].

Two moduli spaces occur. For four marked points labelled $0, \infty, 1, x$, and for five points labelled $0, \infty, 1, y, z$, we use respectively

$$Y = \mathbb{C} \setminus \{0,1\}, \qquad M = \{(y,z) \in Y^2 : y \neq z\}. \tag{4.1}$$

Thus for $A \supset \{0, \infty, 1\}$ with $|A| = 4$, $\pi_A \colon \mathcal{T}(A) \to Y$ records the fourth point under the marking normalized to fix $0, \infty, 1$, and similarly $\pi_B \colon \mathcal{T}(B) \to M$ when $|B| = 5$.

> **Reviewer 3**
>
> Poor choice of notation for the moduli spaces.

Products of based loops are read from right to left: $\alpha\beta$ traverses β first and then α, matching composition of homeomorphisms. For a closed loop ω in $\mathbb{C} \setminus \{c\}$ we write $\mathrm{wind}_c(\omega) \in \mathbb{Z}$ for its winding number about c; for a non-closed path, $\mathrm{wind}_c(\omega)$ denotes the total change of argument of $\omega - c$ divided by 2π, which is additive under concatenation and changes sign under reversal.

Let δ be a loop in $\mathcal{M}(B)$ based at $\pi_B(\tau_0)$, that is, a motion of the marked points. By isotopy extension there is an ambient isotopy h_t of $\mathbb{P}^1$ with $h_0 = \mathrm{id}$ such that $h_t|_B$ follows $\delta(t)$. The mapping class $P(\delta) = [h_1] \in \mathrm{PMod}(B)$ depends only on the homotopy class of δ, and $[h_t]$ is the lift of δ to $\mathcal{T}(B)$ starting at τ_0, with endpoint $\tau_0 \cdot P(\delta)$. Consequently the deck transformation associated with δ, the one sending τ_0 to the endpoint of this lift, is

$$[\phi] \longmapsto [\phi \circ P(\delta)]. \tag{4.2}$$

Point pushing respects products, $P(\alpha\beta) = P(\alpha)P(\beta)$, since the isotopy following $\alpha\beta$ may be taken to be h_t^{β} followed by $h_t^{\alpha} \circ h_1^{\beta}$. Because (4.2) is a right action, deck transformations of products

compose in the opposite order; we only use deck transformations of powers of a single loop. We write $g(\tau)$ for the deck transformation (4.2) of a loop g applied to τ.

When only one marked point v moves, δ is a loop ζ in $\mathbb{P}^1 \setminus (B \setminus \{v\})$ based at v, and $P(\zeta)$ is the *point-pushing map* $\mathrm{Push}(\zeta)$ of the Birman exact sequence [FM, §4.2, Thm. 4.6]. We use two standard facts. (P1) If ζ is a simple loop, then $\mathrm{Push}(\zeta) = T_{\zeta_1} T_{\zeta_2}^{-1}$, where ζ_1, ζ_2 are the two boundary curves of a regular neighbourhood of ζ, the order being fixed by the orientation of ζ and the sign convention for twists [FM, Fact 4.7]. (P2) For every mapping class h and simple closed curve c, $hT_c h^{-1} = T_{h(c)}$ [FM, Fact 3.7].

> **Reviewer 3**
>
> Having two notations $P(\zeta) = \mathrm{Push}(\zeta)$ being used simulatenously is confusing and unnecessary. May generate confusion in section 8 onwards.

Two consequences of (P1) are used. First, if $\Delta \subset \mathbb{P}^1$ is a closed disk with $\partial\Delta \cap B = \emptyset$ and $\Delta \cap B = \{v, w\}$, and ζ is a simple loop of v in $\Delta \setminus \{w\}$ going once around w, then one boundary curve of its regular neighbourhood bounds a disk containing only w (trivial twist) and the other is isotopic to $\partial\Delta$, so

$$\mathrm{Push}(\zeta) = T_{\partial\Delta}^{\pm 1}. \tag{4.3}$$

Second, if κ, η are disjoint simple closed curves bounding an annulus meeting B only in v, with κ inside η in a given disk picture, and ζ is a simple loop of v once counterclockwise around the core of the annulus, then $\mathrm{Push}(\zeta) = T_\eta^{\epsilon} T_\kappa^{-\epsilon}$ with $\epsilon = \pm 1$ depending only on the sign convention for Dehn twists.

Convention 4.1. We fix the sign convention for Dehn twists so that $\epsilon = +1$, that is, $\mathrm{Push}(\zeta) = T_\eta T_\kappa^{-1}$ for such a counterclockwise circuit. With the opposite convention the boxed formula (5.5) of Section 5 reads $T_\eta^{-2}(C)$; both curves have the stated property (Remark 9.4).

5 The model map and the curve

Let f be as in Section 3.2 and conjugate it so that its critical cycle is $0 \mapsto \infty \mapsto 1 \mapsto s \mapsto 0$ with 0 the critical point. A quadratic rational map with a critical point at 0 mapping to ∞ has a double pole at 0 and no other pole, so it is $1 + b_s/z + c_s/z^2$ after using $F(\infty) = 1$. Solving the linear equations $F(1) = s$ and $F(s) = 0$ for b_s, c_s gives $b_s = (-s^2 - s + 1)/(s - 1)$ and $c_s = s(2s - 1)/(s - 1)$, which is (3.3). The other critical point of F_s is $-2c_s/b_s$, and its critical value is $1 - b_s^2/(4c_s)$, which simplifies to

$$v(s) = -\frac{(s^2 - 3s + 1)^2}{4s(s - 1)(2s - 1)}. \tag{5.1}$$

The map has degree two and the four cycle points are distinct exactly when $s \neq 0, 1, 1/2$ (for $s = 1/2$ one has $c_s = 0$). The identities

$$v(s) - 1 = -\frac{(s^2 + s - 1)^2}{4s(s - 1)(2s - 1)}, \qquad v(s) - s = -\frac{(3s^2 - 3s + 1)^2}{4s(s - 1)(2s - 1)}$$

show that $v(s) \notin \{0, \infty, 1, s\}$, that is, $v(s)$ lies outside the cycle, exactly when the three quadratic factors below are nonzero. So the maps in the problem are, up to Möbius conjugacy, precisely the F_s with

$$s \in U = \left\{ s \in \mathbb{C} : s(s - 1)(2s - 1)(s^2 - 3s + 1)(s^2 + s - 1)(3s^2 - 3s + 1) \neq 0 \right\}. \tag{5.2}$$

The set U, a cofinite subset of $\mathbb{C}$, is connected. We write $A_s = \{0, \infty, 1, s\}$ and $B_s = A_s \cup \{v(s)\}$ for the marked sets of F_s. Our model is the parameter $a = 3/5$:

$$F_a(z) = 1 - \frac{1}{10z} - \frac{3}{10z^2}, \qquad 0 \longmapsto \infty \longmapsto 1 \longmapsto a \longmapsto 0, \tag{5.3}$$

with other critical point -6 and other critical value $v_a = v(3/5) = 121/120$. On the sphere marked by $B_a = \{0, \infty, 1, a, v_a\}$ draw the round circles

$$C = \left\{|z| = \tfrac{241}{240}\right\}, \qquad \eta = \left\{\left|z - \tfrac{4}{5}\right| = \tfrac{1}{4}\right\}. \tag{5.4}$$

The exterior of C contains exactly the marked points ∞, v_a (note $v_a = 242/240$), and the interior of η contains exactly $a, 1, v_a$ (the distances from $4/5$ to a, 1, v_a are $1/5$, $1/5$, $25/120$, all less than $1/4$, while $|0 - 4/5| > 1/4$). The two circles intersect. Our model curve is

$$\boxed{\gamma_a = T_\eta^2(C),} \tag{5.5}$$

with the sign of Convention 4.1.

Construction 5.1 (Transport to a general map)**.** *Let f be a map of the family with normalized form F_s, $s \in U$, the normalizing Möbius transformation being μ (so $\mu f \mu^{-1} = F_s$). Choose a path s_t, $0 \le t \le 1$, in U from a to s. The five points $0, \infty, 1, s_t, v(s_t)$ stay distinct, so by isotopy extension there is an ambient isotopy H_t of $\mathbb{P}^1$ with $H_0 = \mathrm{id}$ fixing $0, \infty, 1$ and with $H_t(a) = s_t$, $H_t(v_a) = v(s_t)$. Put $\gamma = \mu^{-1}(H_1(\gamma_a)) \subset \mathbb{P}^1 \setminus B$; equivalently, $H_1(\gamma_a)$ is the image of $H_1(C)$ under two twists about $H_1(\eta)$.*

The isotopy class of γ relative to B depends on the path s_t only through its homotopy class in U; Theorem 5.2 holds for every choice.

Theorem 5.2. *Let f, A, B be as in Section 3.2, and let γ be the curve of Construction 5.1. Then the Dehn twist T_γ preserves $\mathcal{D}_f$. If D_0 denotes the connected component of $\mathcal{D}_f$ containing the base point τ_0, then the components $T_\gamma^n(D_0)$, $n \in \mathbb{Z}$, are pairwise distinct.*

Sections 6–9 prove the theorem for the model $f = F_a$ and $\gamma = \gamma_a$; Section 10 deduces the general case.

6 A finite cover of the moduli space

Recall Y and M from (4.1); M is the moduli space of B_a, with $0, \infty, 1$ normalized and (y, z) the positions of the points corresponding to a and v_a. A point of $\mathcal{T}(B_a)$ determines, through (3.1), a quadratic rational map with marked source and target points; the normalized positions of the fourth source and target points give a map from $\mathcal{T}(B_a)$ to a surface $\mathcal{W}$ of marked maps, a finite cover of M. This is the parameter surface of Hironaka and Koch [HKa, §§4.1–4.2]; in general such spaces are Hurwitz spaces, finite coverings of the moduli space of the target [Ko, Thm. 2.6], [KPS, §3.1].

Consider a quadratic rational map with marked source points $0, \infty, 1, x$ and marked target points $0, \infty, 1, y, z$ realizing the portrait

$$0 \overset{2}{\longmapsto} \infty, \qquad \infty \longmapsto 1, \qquad 1 \longmapsto y, \qquad x \longmapsto 0, \tag{6.1}$$

where the 2 indicates a critical point, and having z as its other critical value. A degree-two map with a double pole at 0 and value 1 at ∞ is $(t-x)(t-r)/t^2$ with zeros x and r, and the condition $1 \mapsto y$ determines r. Hence the map is unique:

$$F_{x,y}(t) = \frac{(t-x)(t-r)}{t^2}, \qquad r = \frac{x+y-1}{x-1}. \tag{6.2}$$

Its critical points are 0 and $2xr/(x+r)$, and the critical value at the latter is $-(x-r)^2/(4xr)$, which in terms of x, y reads

$$Z(x,y) = \frac{((x-1)^2-y)^2}{4x(x+y-1)(1-x)}. \tag{6.3}$$

The marked points must be distinct and the map must have degree two ($r \neq 0$), so the parameter surface and its projection to M are

$$\mathcal{W} = \{(x,y) \in Y^2 : x+y \neq 1,\ Z(x,y) \notin \{0,1,y\}\}, \tag{6.4}$$

$$p\colon \mathcal{W} \longrightarrow M, \qquad p(x,y) = (y, Z(x,y)). \tag{6.5}$$

The maps F_s of the family (3.3) are the marked maps whose fourth source and target points agree: indeed $F_{s,s} = F_s$, the two expressions being quadratic maps with the same poles, zeros and value at ∞, and $Z(s,s) = v(s)$. The conditions $Z(s,s) \neq 0, 1, s$ are precisely the three quadratic factors in (5.2), so the *diagonal* of $\mathcal{W}$ is

$$\mathcal{V} = \mathcal{W} \cap \{x = y\} = \{(s,s) : s \in U\}, \tag{6.6}$$

a connected Riemann surface, since U is connected.

Lemma 6.1. *1. The map p is an unramified covering of degree four, and $\mathcal{W}$ is connected.*

2. For the model (5.3), with $\pi_A\colon \mathcal{T}(A_a) \to Y$ and $\pi_B\colon \mathcal{T}(B_a) \to M$ the projections of Section 4, the map

$$q = (\pi_A \circ F_a^*,\ \pi_A \circ i_*)\colon \mathcal{T}(B_a) \longrightarrow \mathcal{W}$$

is well defined and satisfies $p \circ q = \pi_B$ and $q(\tau_0) = (a,a)$, and it is the universal covering of $\mathcal{W}$. Thus π_B factors through $\mathcal{W}$, and, with $\mathrm{pr}_x, \mathrm{pr}_y$ the two coordinate projections of $\mathcal{W}$, by construction

$$\pi_A \circ F_a^* = \mathrm{pr}_x \circ q, \qquad \pi_A \circ i_* = \mathrm{pr}_y \circ q. \tag{6.7}$$

Proof. (1) Fix $(y,z) \in M$. The equation $Z(x,y) = z$ is

$$((x-1)^2-y)^2 + 4zx(x-1)(x+y-1) = 0,$$

a monic polynomial equation of degree four in x. Its values at $x = 0, 1, 1-y$ are $(1-y)^2$, y^2 and $y^2(y-1)^2$, all nonzero, so every root satisfies $x \in Y$ and $x+y \neq 1$; since $z \notin \{0,1,y\}$, every root gives a point of $\mathcal{W}$. Thus $p^{-1}(y,z)$ consists of the roots of this equation. The roots are distinct because of the factorizations

$$Z - 1 = -\frac{(x^2+y-1)^2}{4x(x-1)(x+y-1)},$$

$$Z - y = -\frac{(x^2+2xy-2x-y+1)^2}{4x(x-1)(x+y-1)},$$

$$\frac{\partial Z}{\partial x} = -\frac{(x^2+y-1)\,((x-1)^2-y)\,(x^2+2xy-2x-y+1)}{4x^2(x-1)^2(x+y-1)^2},$$

which show that $\partial Z/\partial x$ vanishes only where $Z = 1$, $Z = 0$ or $Z = y$, and these values of z are excluded in M. So p has exactly four preimages over each point, $\partial Z/\partial x \neq 0$ on $\mathcal{W}$, and the implicit function theorem shows that p is a local homeomorphism. A local homeomorphism between Hausdorff spaces all of whose fibres have the same finite cardinality is a covering (intersect the images of disjoint neighbourhoods of the four preimages of a point, and count), so p is a covering of degree four. Finally $\mathcal{W}$ is the complement in $\mathbb{C}^2$ of the finitely many algebraic curves listed in (6.4), and such a complement is path connected: join two points by two complex lines through a generic auxiliary point, each meeting the curves in finitely many points, which a path in the line can avoid.

(2) Let $\tau = [\phi] \in \mathcal{T}(B_a)$, with ϕ normalized to fix $0, \infty, 1$, and let $F \circ \psi = \phi \circ F_a$ be a pullback diagram (3.1) with ψ also normalized to fix $0, \infty, 1$ (possible since ψ is unique up to a Möbius transformation). Put $x = \psi(a)$, $y = \phi(a)$, $z = \phi(v_a)$. Applying the diagram to the points of A_a and to the critical points $0, -6$ of F_a shows that F has the portrait (6.1) with marked source points $\{0, \infty, 1, x\}$, marked target points $\{0, \infty, 1, y, z\}$, and other critical value $F(\psi(-6)) = z$. Hence $F = F_{x,y}$ and $Z(x, y) = z$ by the uniqueness statement above; $x, y \in Y$ and $z \notin \{0, 1, y\}$ because ψ, ϕ are injective, and $x + y \neq 1$ because F has degree two. Therefore $(x, y) = (\pi_A F_a^* \tau, \pi_A i_* \tau) \in \mathcal{W}$ and $p(x, y) = (y, z) = \pi_B(\tau)$. For τ_0 we may take $\psi = \phi = \mathrm{id}$, so $q(\tau_0) = (a, a)$; and q is continuous because F_a^*, i_*, π_A are.

It remains to see that q is a covering. Let $N \subsetneq M$ be an open ball evenly covered by both p and π_B, with $p^{-1}(N) = \bigsqcup_j N_j$ and $\pi_B^{-1}(N) = \bigsqcup_k \widetilde{N}_k$. The connected set $q(\widetilde{N}_k)$ lies in a single N_j, where $q|_{\widetilde{N}_k} = (p|_{N_j})^{-1} \circ \pi_B|_{\widetilde{N}_k}$ is a homeomorphism onto N_j. Hence each N_j is evenly covered by q (possibly with empty preimage), so $q(\mathcal{T}(B_a))$ is open and closed in the connected space $\mathcal{W}$, and q is a surjective covering; as $\mathcal{T}(B_a)$ is simply connected, it is the universal covering. □

> **Reviewer 3**
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> Exposition would benefit greatly from having diagrams expressing all the covering and mapping relations between the Teichmuller, moduli, and Hurwitz spaces. Moreover, all of these computations are present in (Hironaka, Koch); not necessarily detrimental to exposition if the goal is to be self-contained, since the work is properly cited.

7 When a loop preserves a component

We now set up the covering-space criterion of Hironaka and Koch [HK, §§2.8–2.11] in the coordinates of Lemma 6.1. Let $b = (a, a) = q(\tau_0)$. A loop g in $\mathcal{W}$ based at b determines, via the loop $p \circ g$ in M and (4.2), a deck transformation δ of π_B, which sends τ_0 to the endpoint τ_1 of the lift of g to $\mathcal{T}(B_a)$ starting at τ_0. It is also a deck transformation of q: $q \circ \delta$ and q are both lifts of π_B through p, and they agree at τ_0 because $q(\tau_1) = b$. We denote δ by the same letter g and write $g(\tau)$, as in Section 4. The two *coordinate loops* of g are the loops $\mathrm{pr}_x \circ g$ and $\mathrm{pr}_y \circ g$ in Y based at a.

Lemma 7.1. *Let $E = \mathrm{im}\big(\pi_1(\mathcal{V}, b) \to \pi_1(\mathcal{W}, b)\big)$ be the set of classes of loops in $\mathcal{W}$ that are homotopic in $\mathcal{W}$ to a loop in the diagonal $\mathcal{V}$. Let $g \in \pi_1(\mathcal{W}, b)$, acting on $\mathcal{T}(B_a)$ as above, and let D_0 be the connected component of $\mathcal{D}_{F_a}$ containing τ_0.*

1. *If the two coordinate loops of g have equal classes in $\pi_1(Y, a)$, then $g(\mathcal{D}_{F_a}) = \mathcal{D}_{F_a}$.*
2. *$g(D_0) = D_0$ if and only if $g \in E$.*

Consequently, if the coordinate loops of g are homotopic and $g^n \notin E$ for every $n \neq 0$, then g preserves $\mathcal{D}_{F_a}$ and the components $g^n(D_0)$, $n \in \mathbb{Z}$, are pairwise distinct.

Proof. (1) By (6.7), F_a^* and i_* are lifts through the universal covering $\pi_A \colon \mathcal{T}(A_a) \to Y$ of the maps $\mathrm{pr}_x \circ q$ and $\mathrm{pr}_y \circ q$, and they take the same value at τ_0 because $\tau_0 \in \mathcal{D}_{F_a}$. Since g is a deck transformation of q, the map $F_a^* \circ g$ is another lift of $\mathrm{pr}_x \circ q$, so $F_a^* \circ g = \delta_x \circ F_a^*$ for a deck transformation δ_x of π_A. To identify δ_x, lift g to a path $\tilde{g}$ from τ_0 to $g(\tau_0)$; then $F_a^* \circ \tilde{g}$ is the lift, starting at $F_a^*(\tau_0)$, of the loop $\mathrm{pr}_x \circ g$, so δ_x is the deck transformation of π_A determined by the class of $\mathrm{pr}_x \circ g$ relative to the base point $F_a^*(\tau_0)$. Likewise $i_* \circ g = \delta_y \circ i_*$ with δ_y determined by the class of $\mathrm{pr}_y \circ g$ relative to the same base point $i_*(\tau_0) = F_a^*(\tau_0)$. Equal classes give $\delta_x = \delta_y =: \delta$, and then for $\tau \in \mathcal{D}_{F_a}$,

$$F_a^*(g\tau) = \delta F_a^*(\tau) = \delta\, i_*(\tau) = i_*(g\tau),$$

so $g(\mathcal{D}_{F_a}) \subset \mathcal{D}_{F_a}$. The coordinate loops of g^{-1} are the inverses of those of g, hence also homotopic, so $g^{-1}(\mathcal{D}_{F_a}) \subset \mathcal{D}_{F_a}$ as well.

(2) First, $\mathcal{D}_{F_a} \subset q^{-1}(\mathcal{V})$: if $F_a^*\tau = i_*\tau$, the two coordinates of $q(\tau)$ agree by (6.7). Next, on a connected component N of $q^{-1}(\mathcal{V})$, the maps $F_a^*|_N$ and $i_*|_N$ are lifts through π_A of one and the same map $\tau \mapsto \mathrm{pr}_x\, q(\tau) = \mathrm{pr}_y\, q(\tau)$, so by uniqueness of lifts they agree on all of N if they agree at one point. Hence $\mathcal{D}_{F_a}$ is a union of connected components of $q^{-1}(\mathcal{V})$; these are path connected, $q \colon q^{-1}(\mathcal{V}) \to \mathcal{V}$ being a covering of the connected manifold $\mathcal{V}$, and D_0 is the component through τ_0.

The deck transformation g permutes the components of $q^{-1}(\mathcal{V})$, so $g(D_0) = D_0$ if and only if there is a path in D_0 from τ_0 to $g(\tau_0)$. Given such a path, its image under q is a loop in $\mathcal{V}$ based at b, homotopic in $\mathcal{W}$ to g because both lift to paths in the simply connected space $\mathcal{T}(B_a)$ with the same endpoints; so $g \in E$. Conversely, if g is homotopic in $\mathcal{W}$ to a loop g' in $\mathcal{V}$, the lift of g' starting at τ_0 stays in $q^{-1}(\mathcal{V})$, hence in D_0, and ends at $g(\tau_0)$ by homotopy lifting; so $g(\tau_0) \in D_0$.

Finally, if $g^j(D_0) = g^k(D_0)$ with $j \neq k$, then $g^{j-k}(D_0) = D_0$, so $g^{j-k} \in E$ by (2), contrary to hypothesis. □

8 Loops in $\mathcal{W}$ representing the twist about γ_a

The goal of this section is a loop g in $\mathcal{W}$ with $P(p \circ g) = T_{\gamma_a}^{\pm 1}$ and trivial coordinate loops (Proposition 8.3). We work in the fibre $\mathcal{W}_a = \{x \in \mathbb{C} : (x, a) \in \mathcal{W}\}$ over $y = a = 3/5$, along which p fixes the four points of A_a and moves only v_a, so that mapping classes are point pushes of v_a. By (6.4) and the factorizations in the proof of Lemma 6.1, the points of $\mathbb{C}$ excluded from $\mathcal{W}_a$ are

$$0,\ 1,\ \tfrac{2}{5}, \qquad 1 \pm \sqrt{\tfrac{3}{5}}\ \ (Z = 0), \qquad \pm\sqrt{\tfrac{2}{5}}\ \ (Z = 1), \qquad \tfrac{2}{5} \pm \tfrac{i\sqrt{6}}{5}\ \ (Z = a); \tag{8.1}$$

here $2/5 = 1 - a$ is the pole of $Z(\cdot, a)$. Their moduli are 0, 1, 0.4, $1.77\ldots$, $0.22\ldots$, and $\sqrt{2/5} = 0.632\ldots$ for the last four; none is $a = 0.6$, so $\ell(\theta) = (ae^{2\pi i\theta}, a)$, $0 \leq \theta \leq 1$, is a loop in $\mathcal{W}$ based at b. Its image $p \circ \ell$ in M moves v_a along the loop $\zeta(\theta) = Z(ae^{2\pi i\theta}, a)$ in $\mathbb{P}^1 \setminus A_a$ based at $v_a = Z(a, a)$.

Let m be a positive meridian about the pole $x = 2/5$ in $\mathcal{W}_a$: travel from a along the real interval toward $2/5$, go once counterclockwise around a small circle centred at $2/5$, and return along the interval.

Lemma 8.1. *The mapping class $P(p \circ m)$ is $T_C^{\pm 1}$, a single Dehn twist about the round circle C of* (5.4).

Proof. On $(2/5, 3/5]$ the factorizations give $\partial Z(x,a)/\partial x < 0$, because in the numerator of $\partial Z/\partial x$ the factors $x^2 - 2/5$ and $(x-1)^2 - 3/5$ are negative and $x^2 - \frac{4}{5}x + \frac{2}{5}$ is positive. Moreover $Z(a,a) = v_a$, so $Z(x,a) \geq v_a > 1$ on $(2/5, a]$ by monotonicity; the pole of $Z(\cdot, a)$ at $2/5$ is simple, and $Z(x,a) \to +\infty$ as $x \downarrow 2/5$. Thus $p \circ m$ moves v_a out along the positive real ray, once around ∞ (a small circle about a simple pole maps to a loop winding once about ∞), and back. For a small enough circle this motion takes place in the closed disk $\Delta = \{|z| \geq 241/240\} \cup \{\infty\}$ bounded by C, whose only marked points are v_a and ∞; in $\Delta \setminus \{\infty\}$, whose fundamental group is $\mathbb{Z}$, the loop is homotopic to a simple loop once around ∞, so (4.3) gives $P(p \circ m) = T_C^{\pm 1}$. □

> **Reviewer 3**
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> Proof could likely be simplified; just need the main takeaway that the image is a simple loop once around ∞.

The homotopy class of ζ, not merely its homology class, must be identified, since the isotopy class of γ_a depends on it.

Lemma 8.2. *The loop ζ is homotopic, as a loop based at v_a in $\mathbb{P}^1 \setminus A_a$, to two clockwise circuits around the segment $[a,1]$ inside the disk bounded by η. Consequently $h := P(\zeta^{-1})$ satisfies $h(C) = T_\eta^2(C) = \gamma_a$, with the sign of Convention 4.1.*

Proof. Step 1: enlarging the circle. Put $R = \sqrt{1-a} = \sqrt{2/5} > a$. No excluded point in (8.1) has modulus in $[a, R)$, so for $a \leq r < R$ the loops $\zeta_r(\theta) = Z(re^{2\pi i\theta}, a)$ avoid $0, a, 1, \infty$ and depend continuously on r. Their base points $Z(r,a)$ move along the real path $\sigma_r = Z([a,r], a) \subset (1, v_a]$, since $Z(\cdot, a)$ is decreasing on $(2/5, R)$ (the sign analysis in the proof of Lemma 8.1 is valid there) with $Z(R,a) = 1$. Hence ζ is based-homotopic in $\mathbb{P}^1 \setminus A_a$ to $\sigma_r^{-1}\zeta_r\sigma_r$ for every $r \in [a, R)$.

Step 2: the circle $|x| = R$ maps onto $[a,1]$, and nothing inside it does. The substitution $u = x + (1-a)/x$ (a Joukowski map) simplifies Z: writing $(x-1)^2 - a = x(u-2)$ and $(x+a-1)(1-x) = x(2-a-u)$ gives

$$Z(x,a) = G(u), \qquad G(u) = \frac{(u-2)^2}{4(2-a-u)}. \tag{8.2}$$

For $x = Re^{i\vartheta}$ we have $u = 2R\cos\vartheta$, so on $|x| = R$ the variable u runs over $[-2R, 2R]$, where $2 - a - u \geq 2 - a - 2R > 0$ and

$$G(u) - 1 = \frac{u^2 - 4(1-a)}{4(2-a-u)}, \qquad G(u) - a = \frac{(u - 2(1-a))^2}{4(2-a-u)}.$$

Thus on $[-2R, 2R]$ the function G takes values in $[a,1]$, equals 1 exactly at the endpoints, and equals a exactly at $u = 2(1-a)$; since $G'(u) = (u-2)(2-2a-u)/(4(2-a-u)^2)$, G decreases from 1 to a and then increases back to 1. Hence for $z \in (a,1)$ the equation $G(u) = z$ has two distinct solutions $u \in (-2R, 2R)$, and for each of them the two roots of $x^2 - ux + (1-a) = 0$ are non-real conjugates with $|x|^2 = 1 - a$, hence lie on $|x| = R$. This produces four distinct points of $|x| = R$ mapping to z, which exhaust the four preimages of (a, z) under p. For the endpoints the factorizations of $Z - 1$ and $Z - a$ show that all preimages of 1 and of a (namely $\pm R$ and $2/5 \pm i\sqrt{6}/5$, with multiplicity) also lie on $|x| = R$. So $Z(\{|x| = R\}, a) = [a,1]$, and no x with $|x| < R$ maps into $[a,1]$.

> **Reviewer 3**
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> Very convoluted writing: could re-emphasize that the domain is still the fiber $\mathcal{W}_a$, and what exactly these coordinate changes are supposed to mean. The conclusion of this paragraph is equally poorly explained.

Step 3: the homotopy class. The segment $[a,1]$ lies in the open disk bounded by η, so by Step 2 and continuity, for $r<R$ close to R the loop ζ_r lies in that disk and avoids $[a,1]$; the base point path $\sigma_r\subset(1,v_a]$ also lies in the disk and avoids $[a,1]$. So $\sigma_r^{-1}\zeta_r\sigma_r$ is a loop in the η-disk minus the segment $[a,1]$. This space is an annulus, with fundamental group $\mathbb{Z}$ generated by a counterclockwise circuit around $[a,1]$, and the class of a loop is its winding number about any point of the segment, say about 1. Since the homotopy of Step 1 avoids 1, this winding number equals $\mathrm{wind}_1(\zeta)$, which we compute by the argument principle applied to

$$Z(x,a)-1=-\frac{(x^2+a-1)^2}{4x(x-1)(x+a-1)}$$

on the disk $|x|<a$: the numerator vanishes only at $\pm R$, outside the disk, while the denominator has the two simple zeros 0 and $1-a=2/5$ inside it. Hence $\mathrm{wind}_1(\zeta)=0-2=-2$, and ζ is based-homotopic to two clockwise circuits around $[a,1]$ in the η-disk, as claimed.

Step 4: the point push. Choose a round circle κ enclosing $a,1$ but not v_a and lying inside both C and η; for instance $\kappa=\{|z-\frac{4}{5}|=\frac{97}{480}\}$, for which $|a-\frac{4}{5}|=|1-\frac{4}{5}|=\frac{96}{480}<\frac{97}{480}<\frac{100}{480}=|v_a-\frac{4}{5}|$, and the largest modulus on κ is $\frac{481}{480}<\frac{241}{240}$. The annulus between κ and η contains the single marked point v_a, and by Step 3, ζ^{-1} is homotopic to two counterclockwise circuits of v_a around this annulus. By Convention 4.1 one such circuit has point push $T_\eta T_\kappa^{-1}$, so, the twists about the disjoint curves κ and η commuting,

$$h=P(\zeta^{-1})=T_\eta^2T_\kappa^{-2}.$$

Since κ is disjoint from C, the twist T_κ fixes the isotopy class of C, and $h(C)=T_\eta^2(C)=\gamma_a$. □

Proposition 8.3. *Let*

$$g=\ell^{-1}m\ell\in\pi_1(\mathcal{W},b),\tag{8.3}$$

the loop that traverses ℓ, then m, then ℓ backwards. Then $P(p\circ g)=T_{\gamma_a}^{\pm1}$: the deck transformation g acts on $\mathcal{T}(B_a)$ as the Dehn twist about γ_a or its inverse. Both coordinate loops of g are null-homotopic in Y. In particular T_{γ_a} preserves $\mathcal{D}_{F_a}$.

Proof. Since $p\circ\ell$ moves only v_a, along ζ, we have $P(p\circ\ell)=P(\zeta)=h^{-1}$ with h as in Lemma 8.2. Multiplicativity of P (Section 4) and Lemma 8.1 give

$$P(p\circ g)=P(p\circ\ell^{-1})\,P(p\circ m)\,P(p\circ\ell)=h\,T_C^{\pm1}\,h^{-1}=T_{h(C)}^{\pm1}=T_{\gamma_a}^{\pm1},$$

using (P2) and Lemma 8.2. The y-coordinate of g is constant. The x-coordinate of m is a small loop around $2/5$ together with a real approach path in $(2/5,3/5]$; this loop lies in the disk $|x-\frac{1}{2}|<\frac{1}{2}$, which avoids 0 and 1, so it is null-homotopic in Y; hence the x-coordinate loop of g is homotopic to $(\mathrm{pr}_x\circ\ell)^{-1}(\mathrm{pr}_x\circ\ell)$, which is trivial. Lemma 7.1(1) now shows that g, hence T_{γ_a}, preserves $\mathcal{D}_{F_a}$. □

> **Reviewer 3**
>
> Section has poor exposition, stemming from poor choice of notation, lack of any diagrams or drawings in assisting the visualization of the proof or proof strategy, and overall overly detailed characterizations

of the loops involved.

9 No nonzero power of the twist preserves the base component

By Lemma 7.1 and Proposition 8.3, it remains to show that $g^n \notin E$ for $n \neq 0$. Consider

$$\Phi\colon \mathcal{W} \longrightarrow Y \times \mathbb{C}^*, \qquad \Phi(x,y) = \big(t(x,y), L(x,y)\big) = \left(\frac{xy}{(1-x)(1-y)},\ (1-x)(1-y)\right). \tag{9.1}$$

It is well defined: $L \neq 0$ and $t \neq 0$ since $x, y \in Y$, and $t - 1 = (x+y-1)/((1-x)(1-y)) \neq 0$. In terms of the symmetric functions $e_1 = x+y$, $e_2 = xy$ one has $L = 1 - e_1 + e_2$ and $t = e_2/L$, so Φ is the symmetric quotient $(x,y) \mapsto (x+y, xy)$ of Hironaka–Koch [HK, §2.12], restricted to $\mathcal{W}$, up to this change of coordinates.

The constraint on diagonal loops. Enlarge the diagonal by filling in its six quadratic punctures: $\widehat{\mathcal{V}} = \{(s,s) : s \in \mathbb{C} \setminus \{0, 1, 1/2\}\} \supset \mathcal{V}$, on which (9.1) still makes sense, so that the covering (9.3) below covers all of Y. In the coordinate $w = s/(1-s)$, which maps $\mathbb{C} \setminus \{0, 1, 1/2\}$ bijectively onto $\mathbb{C} \setminus \{0, 1, -1\}$, the restriction of Φ is

$$t = w^2, \qquad L = (1+w)^{-2}, \qquad w \in \mathbb{C} \setminus \{0, 1, -1\}, \tag{9.2}$$

because $1 + w = 1/(1-s)$. The squaring map

$$\mathbb{C} \setminus \{0, 1, -1\} \longrightarrow Y, \qquad w \longmapsto w^2, \tag{9.3}$$

is a two-sheeted covering; our base points are $w_0 = a/(1-a) = 3/2$ and $t_0 = w_0^2 = 9/4$.

Reviewer 3

Non-obvious coordinate changes; seems arcane at first but provides a somewhat simplification of the arguments of the human solution and of the symmetric quotient construction of (Hironaka, Koch).

Lemma 9.1. *Let λ be a loop in $\mathcal{V}$ based at b. Then $w \circ \lambda$ is the lift of the loop $t \circ \lambda$ through* (9.3) *starting at w_0; it is closed, and*

$$\operatorname{wind}_0(L \circ \lambda) = -2 \operatorname{wind}_{-1}(w \circ \lambda). \tag{9.4}$$

The based homotopy class of $w \circ \lambda$ in $\mathbb{C} \setminus \{0, 1, -1\}$, hence the right-hand side of (9.4)*, depends only on the based homotopy class of $t \circ \lambda$ in Y.*

Proof. On $\mathcal{V} \subset \widehat{\mathcal{V}}$ formula (9.2) applies: $w \circ \lambda$ is a closed path in $\mathbb{C} \setminus \{0, 1, -1\}$ from w_0 with $(w \circ \lambda)^2 = t \circ \lambda$, hence the lift of $t \circ \lambda$ from w_0, and $L \circ \lambda = (1 + w \circ \lambda)^{-2}$ gives (9.4). The last assertion is homotopy lifting for the covering (9.3). □

The loops g^n violate the constraint. On the fibre $y = a$ the map (9.1) becomes

$$t = \frac{3x}{2(1-x)}, \qquad L = \frac{2}{5}(1-x). \tag{9.5}$$

Let $\alpha = t \circ \ell$ and $\beta = t \circ m$, loops in Y based at t_0.

Lemma 9.2. *For every integer n, the loop g^n satisfies*

$$\operatorname{wind}_0(L \circ g^n) = 0, \qquad \operatorname{wind}_{-1}\big(\widetilde{t \circ g^n}\big) = n, \tag{9.6}$$

where $\widetilde{t \circ g^n}$ is the lift of $t \circ g^n$ through (9.3) starting at w_0, which is a closed loop.

Proof. The loop ℓ. Along ℓ we have $x = ae^{2\pi i\theta}$ with $|x| = a < 1$, so $1 - x$ does not wind about 0 and (9.5) gives $\operatorname{wind}_0(\alpha) = 1$ and $\operatorname{wind}_0(L \circ \ell) = 0$. Since $\operatorname{wind}_0(\alpha)$ is odd, continuation of the square root along α, or along α^{-1}, interchanges the two sheets of (9.3): the lift $\tilde{\alpha}$ of α starting at w_0 ends at $-w_0$.

The loop m. The Möbius map $x \mapsto t$ in (9.5) sends $2/5$ to 1 and a to $9/4$, is increasing on the real interval $(2/5, a]$, which it maps onto $(1, 9/4]$, and, being conformal, sends a small counterclockwise circle about $2/5$ to a small counterclockwise loop about 1. Hence β is a positive meridian about 1 in Y, approached from t_0 along the real axis to the right of 1. Its lift starting at $w_0 = 3/2$ runs along $(1, 3/2]$ and around a small loop about $+1$, so it is a positive meridian about $+1$; the lift starting at $-w_0$ is its negative, a positive meridian about -1 (the map $w \mapsto -w$ preserves orientation). Also $\operatorname{wind}_0(L \circ m) = 0$, since $1 - x$ stays near the positive interval $[2/5, 3/5]$ along m.

The loop g. By additivity, $\operatorname{wind}_0(L \circ g) = -\operatorname{wind}_0(L \circ \ell) + \operatorname{wind}_0(L \circ m) + \operatorname{wind}_0(L \circ \ell) = 0$, and so $\operatorname{wind}_0(L \circ g^n) = 0$. The t-projection of g is $\alpha^{-1}\beta\alpha$: first α, then β, then α backwards. Its lift from w_0 consists of $\tilde{\alpha}$, ending on the other sheet at $-w_0$, then the lift of β from $-w_0$, a positive meridian about -1, then $\tilde{\alpha}$ reversed, returning to w_0. The contributions of $\tilde{\alpha}$ and its reverse to the change of argument of $w + 1$ cancel, so this closed lift has winding number 1 about -1, and the lift of $t \circ g^n = (\alpha^{-1}\beta\alpha)^n$ from w_0, the n-fold concatenation, has winding number n. □

Proposition 9.3. *For every $n \neq 0$, $g^n \notin E$. Consequently Theorem 5.2 holds for the model map F_a and the curve $\gamma_a = T_\eta^2(C)$.*

Proof. Suppose $g^n \in E$, so that g^n is homotopic in $\mathcal{W}$ to a loop λ in $\mathcal{V}$ based at b. Applying Φ, the loops $t \circ \lambda$ and $t \circ g^n$ are based-homotopic in Y, and $L \circ \lambda$ and $L \circ g^n$ are homotopic in $\mathbb{C}^*$. By Lemma 9.1, the lift $w \circ \lambda$ is based-homotopic to the lift of $t \circ g^n$ used in Lemma 9.2, so $\operatorname{wind}_{-1}(w \circ \lambda) = n$, while $\operatorname{wind}_0(L \circ \lambda) = \operatorname{wind}_0(L \circ g^n) = 0$. The constraint (9.4) then reads $0 = -2n$, so $n = 0$.

For the second assertion, g acts as $T_{\gamma_a}^{\pm 1}$ and has homotopic coordinate loops (Proposition 8.3), so by Lemma 7.1 the twist T_{γ_a} preserves $\mathcal{D}_{F_a}$ and the components $T_{\gamma_a}^n(D_0) = g^{\pm n}(D_0)$, $n \in \mathbb{Z}$, are pairwise distinct. □

Remark 9.4. The same computation applies to the other conjugate $g' = \ell m \ell^{-1}$ (its t-projection $\alpha\beta\alpha^{-1}$ also changes sheet before the meridian, as α^{-1} has odd winding number about 0, and again $\operatorname{wind}_0(L \circ g') = 0$), whose mapping class is $h^{-1}T_C^{\pm 1}h = T_{h^{-1}(C)}^{\pm 1}$ with $h^{-1}(C) = T_\eta^{-2}T_\kappa^2(C) = T_\eta^{-2}(C)$. Hence Theorem 5.2 holds for both curves $T_\eta^{\pm 2}(C)$.

10 Transport to an arbitrary map of the family

Let f, μ, F_s, s_t, H_t be as in Construction 5.1. We show that changing markings by $H = H_1$ carries $\mathcal{D}_{F_a}$ onto $\mathcal{D}_{F_s}$, D_0 to the base component of $\mathcal{D}_{F_s}$, and conjugates T_{γ_a} to $T_{H(\gamma_a)}$.

Lemma 10.1. *There is a continuous family of homeomorphisms K_t of $\mathbb{P}^1$, $0 \le t \le 1$, with $K_0 = \mathrm{id}$ and*

$$F_{s_t} \circ K_t = H_t \circ F_a \qquad (0 \le t \le 1). \tag{10.1}$$

It satisfies $K_t|_{A_a} = H_t|_{A_a}$ for all t; consequently $H_t^{-1}K_t$ is an isotopy from the identity through homeomorphisms fixing A_a pointwise, and K_1 is isotopic to H_1 relative to A_a.

Proof. Every F_s is Möbius-equivalent to $Q(z) = z^2$, with Möbius maps depending continuously on s: completing the square in $F_s(z) = 1 + b_s/z + c_s/z^2$ and recalling $v(s) = 1 - b_s^2/(4c_s)$ gives $F_s = v(s) + c_s\, J_s^2$, that is,

$$\nu_s \circ F_s = Q \circ J_s, \qquad J_s(z) = \frac{1}{z} + \frac{b_s}{2c_s}, \qquad \nu_s(w) = \frac{w - v(s)}{c_s},$$

with $c_s \neq 0$ for $s \in U$. The isotopy $\widehat{H}_t = \nu_{s_t} \circ H_t \circ \nu_a^{-1}$ starts at the identity and fixes 0 and ∞, because $H_t(v_a) = v(s_t)$ and $H_t(\infty) = \infty$. By the homotopy lifting property of the covering $Q\colon \mathbb{C}^* \to \mathbb{C}^*$, the homotopy $(z,t) \mapsto \widehat{H}_t(Q(z))$ lifts uniquely to a continuous $\rho\colon \mathbb{C}^* \times [0,1] \to \mathbb{C}^*$ with $\rho_0 = \mathrm{id}$ and $Q \circ \rho_t = \widehat{H}_t \circ Q$; lifting $\widehat{H}_t^{-1} \circ Q$ in the same way gives ρ_t' with $\rho_t'\rho_t = \mathrm{id} = \rho_t\rho_t'$ (both products lift the constant homotopy Q and start at id), so each ρ_t is a homeomorphism of $\mathbb{C}^*$, and ρ extends continuously to $\mathbb{P}^1 \times [0,1]$ by $\rho_t(0) = 0$, $\rho_t(\infty) = \infty$, since $|\rho_t(z)|^2 = |\widehat{H}_t(z^2)|$ and $\widehat{H}$ is continuous on the compact space $\mathbb{P}^1 \times [0,1]$. Then $K_t = J_{s_t}^{-1} \circ \rho_t \circ J_a$ is a continuous family of homeomorphisms with $K_0 = \mathrm{id}$ and $F_{s_t}K_t = \nu_{s_t}^{-1} Q \rho_t J_a = \nu_{s_t}^{-1}\widehat{H}_t Q J_a = H_t \nu_a^{-1} Q J_a = H_t F_a$, which is (10.1).

For the marked points, (10.1) gives $F_{s_t}(K_t(w)) = H_t(F_a(w))$ for $w \in A_a$. Thus $K_t(0)$ is the critical point of F_{s_t} over ∞, namely 0; $K_t(\infty)$ lies in $F_{s_t}^{-1}(1) = \{\infty, -c_{s_t}/b_{s_t}\}$, $K_t(1)$ in $F_{s_t}^{-1}(s_t)$, and $K_t(a)$ in $F_{s_t}^{-1}(0) \ni s_t$. For $s_t \in U$ each of the latter three fibres (over the non-critical values 1, s_t, 0) consists of two distinct points depending continuously on t: $b_{s_t} \neq 0$ since $s_t^2 + s_t - 1 \neq 0$, and s_t, 0 are not critical values of F_{s_t} since $v(s_t) \neq s_t, 0$. By continuity from $t = 0$, $K_t(0) = 0$, $K_t(\infty) = \infty$, $K_t(1) = 1$ and $K_t(a) = s_t$, which are the values of H_t on A_a. The remaining assertions follow. □

Proof of Theorem 5.2. Write $H = H_1$, $K = K_1$. Since H maps B_a onto B_s (notation of Section 5), changing markings by H is a biholomorphic isomorphism

$$\Psi\colon \mathcal{T}(B_a) \longrightarrow \mathcal{T}(B_s), \qquad [\phi] \longmapsto [\phi \circ H^{-1}],$$

and likewise $[\psi] \mapsto [\psi \circ H^{-1}]$ identifies $\mathcal{T}(A_a)$ with $\mathcal{T}(A_s)$, $A_s = \{0, \infty, 1, s\} = H(A_a) = K(A_a)$.

Ψ carries $\mathcal{D}_{F_a}$ onto $\mathcal{D}_{F_s}$. Let ϕ be a marking with pullback diagram $\widetilde{F} \circ \psi = \phi \circ F_a$ for F_a. Using $F_a \circ K^{-1} = H^{-1} \circ F_s$ from (10.1),

$$\widetilde{F} \circ (\psi K^{-1}) = \phi \circ F_a \circ K^{-1} = (\phi H^{-1}) \circ F_s,$$

so ψK^{-1} is a source marking for the target marking ϕH^{-1} in a pullback diagram for F_s: $F_s^*\Psi[\phi] = [\psi K^{-1}]$, while $i_*\Psi[\phi] = [\phi H^{-1}]$. Now $[\psi K^{-1}] = [\phi H^{-1}]$ in $\mathcal{T}(A_s)$ if and only if $[\psi K^{-1} H] = [\phi]$ in $\mathcal{T}(A_a)$, and $K^{-1}H$ is isotopic to the identity relative to A_a by Lemma 10.1, so this holds if and only if $[\psi] = [\phi]$ in $\mathcal{T}(A_a)$, that is, $[\phi] \in \mathcal{D}_{F_a}$. Hence $\Psi(\mathcal{D}_{F_a}) = \mathcal{D}_{F_s}$.

Ψ conjugates twists. For a mapping class $k \in \mathrm{PMod}(B_a)$, $\Psi([\phi] \cdot k) = [\phi k H^{-1}] = [\phi H^{-1}] \cdot (HkH^{-1})$, so Ψ intertwines the action of T_{γ_a} with that of $HT_{\gamma_a}H^{-1} = T_{H(\gamma_a)}$, by the naturality of twists (P2), which holds verbatim for the homeomorphism $H\colon (\mathbb{P}^1, B_a) \to (\mathbb{P}^1, B_s)$.

Ψ *carries* D_0 *to the base component.* The path of markings $[H_t]$ in $\mathcal{T}(B_a)$ lies in $\mathcal{D}_{F_a}$: by (10.1), K_t is a source marking for the target marking H_t, and $[K_t] = [H_t]$ in $\mathcal{T}(A_a)$ because $H_t^{-1}K_t$ is isotopic to the identity relative to A_a (Lemma 10.1). The path starts at τ_0, so its endpoint $[H]$ lies in D_0, and $\Psi[H] = [\mathrm{id}]$ is the base point of $\mathcal{T}(B_s)$. Hence $\Psi(D_0)$ is the base component of $\mathcal{D}_{F_s}$, and the components $T^n_{H(\gamma_a)}(\Psi(D_0)) = \Psi(T^n_{\gamma_a}(D_0))$, $n \in \mathbb{Z}$, are pairwise distinct by Proposition 9.3. Finally, conjugating by μ^{-1} identifies $(\mathbb{P}^1, B_s, F_s)$ with $(\mathbb{P}^1, B, f)$ and $H(\gamma_a)$ with γ, which proves the theorem for f. □

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